PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

Online Reservation and Management System for Messiah Inland Resort

In partial fulfillment for the requirements in Capstone 1 and Research Subject

Charles Darwen Salcedo BSIT

Elyza Egamino BSIT

John Andrew Romanos BSIT

Lyle Valderrama BSIT

John Cyril Eiman BSIT

Second Semester
2025-26

INTRODUCTION

Technology has become an important part of daily life, especially in how businesses operate and provide services to their customers. In the hospitality industry, resorts are expected to deliver fast, convenient, and reliable services to meet the growing expectations of guests. One of the most important aspects of resort operations is the reservation process, which plays a big role in customer satisfaction and overall efficiency.

Many small and medium-sized resorts still use manual methods such as phone calls, walk-in bookings, and handwritten records. Although these methods may work, they are often time-consuming and prone to errors. Problems such as double bookings, lost records, and delays in responding to customer inquiries can occur. These challenges can affect not only the workflow of the staff but also the experience of the guests.

Messiah Inland Resort is currently using a manual system for handling reservations and managing records. As the number of guests increases, especially during peak seasons, it becomes more difficult for the staff to manage bookings efficiently. This can lead to confusion, miscommunication, and inconvenience for both employees and customers. Because of this, there is a need to improve the current system and make it more reliable and efficient.

With the advancement of technology, online reservation systems have become more common and useful. These systems allow customers to check room availability, make reservations, and receive confirmations anytime and anywhere. For the resort, it helps organize data, reduce errors, and improve overall operations.

This study focuses on developing an Online Reservation and Management System for Messiah Inland Resort. The proposed system aims to simplify the reservation process, provide accurate and real-time information, and help staff manage daily operations more effectively. By implementing this system, the resort can improve service quality, increase customer satisfaction, and keep up with the demands of modern technology.

CHAPTER I

INTRODUCTION

1.1 Background of the Study

Resorts need a reliable way to manage reservations, organize records, and provide good service to their guests. However, many small and medium-sized resorts still depend on manual methods such as phone calls, walk-in bookings, and handwritten logs. While these methods may be manageable at first, they often become difficult to handle as the number of guests increases. Problems like double bookings, missing or incorrect records, slow responses to customer inquiries, and difficulty in checking room availability are common. These issues can lead to confusion and inconvenience for both staff and customers.

Messiah Inland Resort is currently experiencing these kinds of challenges. Since the reservation process is done manually, staff members sometimes struggle to keep track of bookings, especially during busy days or peak seasons. This can result in delays, miscommunication, and occasional errors in reservations. Customers may also find it inconvenient to rely only on calls or personal visits just to check availability or make a booking. Over time, these problems can affect customer satisfaction and may also impact the reputation of the resort.

In today's time, most customers expect faster and more convenient services. Many people prefer using online platforms where they can easily check room availability, make reservations, and receive confirmation instantly. Because of this, resorts that still rely on manual systems may find it harder to keep up with customer expectations. Adapting to a digital system is becoming more important to stay competitive and provide better service.

An online reservation and management system can help solve these problems. It can provide real-time updates on room availability, reduce the chances of double bookings, and store customer information in an organized way. It can also make communication faster and more efficient since customers can book anytime without needing to contact the resort directly. For the staff, it reduces workload and makes daily tasks easier to manage.

Aside from improving the reservation process, the system can also help in monitoring and reporting. The resort can track the number of bookings, identify peak seasons, and understand customer preferences. This information can be useful for planning, improving services, and making better decisions in the future. It also helps ensure that important records are not lost and can be accessed easily when needed.

This study aims to develop an Online Reservation and Management System for Messiah Inland Resort. The system is designed to make the reservation process faster, more accurate, and more convenient for both customers and staff. It will help manage bookings, track room availability, and store important data securely. In the long run, the system is expected to improve the overall operation of the resort, enhance customer satisfaction, and provide a more efficient and organized way of managing reservations.

1.2 Statement of the Problem

Despite the need for an efficient booking system, Messiah Inland Resort still faces several challenges due to its manual reservation process. This study aims to address the following issues:

- Manual reservations can lead to booking conflicts, errors, and confusion in scheduling.
- Customer records are difficult to manage, organize, and may be misplaced or lost over time.
- Staff cannot easily check room and facility availability in real time, which can result in incorrect information being given to customers.
- Handling inquiries and bookings is slow, especially during busy periods or peak seasons.
- There is no automated system to manage reservations and daily operations efficiently.
- The current process relies heavily on human effort, increasing the chances of mistakes and miscommunication.
- There is no centralized database for storing and accessing important customer and reservation information.
- Generating reports, such as daily bookings or customer history, is time-consuming and often inaccurate.

- Customers do not have the option to book online, which can be inconvenient and limit potential reservations.

- The resort has difficulty tracking reservation status, cancellations, and updates effectively.
- Communication between staff and customers is not always fast or consistent.
- The lack of a digital system makes it harder for management to monitor performance and make informed decisions.

These problems highlight the need for a more efficient and reliable system that can improve the reservation process and overall management of the resort.

1.3 Objectives of the Study

1.3.1 General Objective

The main goal of this study is to develop an Online Reservation and Management System for Messiah Inland Resort to make the booking process more efficient, improve record-keeping, and help staff manage daily operations more effectively.

1.3.2 Specific Objectives

Specifically, this study aims to:

- Provide an easy and convenient way for customers to check room and facility availability online.
- Allow customers to make reservations anytime and anywhere without the need for manual inquiries.
- Reduce booking errors such as double reservations and incorrect information.
- Organize and store customer records in a secure and centralized database.
- Enable staff to monitor reservations and availability in real time.
- Speed up the process of handling customer inquiries and bookings.
- Improve communication between the resort staff and customers.
- Generate accurate reports for reservations, customer data, and daily operations.
- Help management track performance and make better decisions based on data.
- Enhance the overall efficiency of the resort's operations.
- Increase customer satisfaction by providing faster and more reliable service.

Through the development of this system, the resort is expected to have a more organized, accurate, and user-friendly way of managing its reservation process.

1.4 Scope and Delimitation

This study focuses on the development of a prototype Online Reservation and Management System for Messiah Inland Resort. The system is designed to improve the current reservation process by providing a more organized, accurate, secure, and efficient way of handling bookings, customer records, and resort operations. The proposed system aims to replace manual reservation procedures with a computerized process that minimizes paperwork, reduces human errors, and improves overall operational efficiency.

The system will include features such as online booking, real-time room and facility availability tracking, customer record management, customer feedback management, and an admin dashboard that allows staff to manage reservations and monitor daily resort activities. It will also allow users to browse available rooms and facilities, check pricing and amenities, make reservations, submit inquiries, and receive booking confirmations. On the admin side, authorized staff can update room availability, manage customer information, monitor reservation status, review customer feedback, and generate reports related to reservations, occupancy, and resort operations.

Additionally, the system will provide a centralized database where all reservation records, customer details, and operational data are securely stored and organized. This centralized approach improves accessibility and helps staff retrieve information quickly and accurately. The system is also expected to improve communication between customers and resort staff by providing updated reservation information and organized transaction records.

The study will cover the planning, design, development, testing, and evaluation of the system to demonstrate its functionality, reliability, and usability. It will focus on improving the reservation workflow, reducing scheduling conflicts, avoiding double bookings, and maintaining accurate room and facility availability. The system will be accessible through a web-based platform to ensure convenience and ease of access for both customers and administrators using internet-connected devices.

Furthermore, the system aims to enhance customer satisfaction by allowing users to conveniently reserve rooms without the need for physical visits or manual coordination. Customers can quickly access resort information, check room availability in real time, and receive immediate confirmation of their reservations. The inclusion of customer feedback functionality will also allow the resort management to evaluate customer experiences and identify areas for service improvement.

The proposed system is also expected to support resort staff by simplifying administrative tasks such as reservation monitoring, report generation, customer management, and availability scheduling. By automating these processes, the staff can perform operations more efficiently and reduce the time spent on manual record keeping.

However, this study has certain limitations. The system will not include large-scale deployment or actual implementation in a live business environment. It will not support third-party payment integration such as online banking, credit card processing, or e-wallet services, and reservations will not involve real financial transactions. Advanced features such as artificial intelligence-based recommendations, detailed business analytics, mobile application support, automated SMS or email notifications, GPS integration, and cloud-based backup services may also be limited or not fully implemented.

In addition, the study will focus only on the core functions of the reservation and management system and will not include advanced security mechanisms used in commercial systems. The prototype may also have limitations in terms of scalability, performance optimization, and multi-branch resort management since it is primarily intended for academic demonstration purposes.

The testing phase of the study will mainly evaluate the system's functionality, usability, reliability, and user acceptance. Feedback gathered during testing may be used to identify system strengths, weaknesses, and possible improvements for future development. Recommendations for future enhancements may include mobile compatibility, online payment integration, automated notification systems, advanced reporting tools, and expanded customer service features.

Overall, despite its limitations, the proposed Online Reservation and Management System aims to provide a practical and efficient solution for improving reservation management and resort operations at Messiah Inland Resort.

1.5 Significance of the Study

The proposed system will benefit the following individuals and groups:

Guests – The system will provide a faster, more convenient, and accurate reservation process. Guests will be able to check room availability, make bookings, and receive confirmation without the need to visit or call the resort, saving both time and effort.

Resort Staff/Administrators – The system will help simplify daily operations such as managing bookings, updating availability, and handling customer records. It will reduce manual work, minimize errors, and allow staff to focus more on providing quality service to guests.

Messiah Inland Resort – The resort will benefit from improved efficiency, better organization of records, and reduced chances of booking conflicts. It will also enhance customer satisfaction, which may lead to better reputation and increased bookings.

BSIT Students – This study will serve as a useful learning tool for students, especially in understanding system development, database management, web-based applications, and real-world problem-solving in the hospitality industry.

Future Researchers – The system can serve as a reference for future studies related to online reservation systems, hotel and resort management systems, or similar web-based applications. It may also help them improve or develop more advanced versions of such systems.

Management/Owners – The system will help management make better decisions through more organized records and easier access to reservation data. It can also help identify booking trends and improve overall business planning.

Local Tourism Industry – By improving the efficiency of resort operations, the system may contribute to better tourism services in the area, encouraging more visitors and supporting local tourism growth.

Overall, the system aims to improve the reservation process, enhance service quality, and support the modernization of resort management.

1.6 DEFINITION OF TERMS

Online Reservation System – A web-based software that allows guests to book rooms, facilities, or services online without needing to visit or call the resort.

Reservation Management – The process of organizing, tracking, updating, and controlling all guest bookings to ensure smooth and accurate operations.

Real-Time Availability – A system feature that instantly shows updated room or facility availability to prevent double bookings and scheduling conflicts.

Customer Records – A digital collection of guest information such as name, contact details, booking history, and preferences, stored securely in the system.

Administrator – Authorized resort staff who manage the system, handle reservations, update availability, and monitor all transactions and records.

System Prototype – An early or initial version of the system developed for testing, evaluation, and demonstration before full implementation.

Database – A structured storage system used to save and manage all information such as bookings, customer details, and room data.

User Interface (UI) – The visual layout of the system that users interact with, including buttons, forms, and menus used for booking and managing reservations.

Booking Confirmation – A notification or record given to customers confirming that their reservation has been successfully made.

Web-Based System – A system that runs through a web browser, allowing users to access it using the internet without installing software.

Accessibility – The ability of users to access and use the system easily at any time and from different devices such as phones, tablets, or computers.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

The rapid advancement of technology has significantly transformed the hospitality and tourism industry, particularly in the area of reservation and booking systems. According to Smith and Anderson (2021), online reservation systems improve operational efficiency by minimizing booking conflicts, reducing human error, and automating reservation processes. Their study emphasized that traditional manual reservation methods are often prone to double bookings, inaccurate records, and delays in customer service. Through automation and real-time updates, online systems help businesses provide faster and more reliable services to customers.

Smith and Anderson further explained that automated reservation systems allow customers to access booking services anytime and anywhere, which increases convenience and accessibility. The researchers highlighted that businesses using digital reservation systems experience improved workflow management because employees can focus on customer relations instead of manual administrative tasks. The study also revealed that online systems enhance transparency since customers can immediately view available schedules, room availability, and booking confirmations in real time.

Similarly, Johnson et al. (2020) conducted a study on automated booking systems and their effects on customer satisfaction in the hospitality sector. Their findings showed that customers prefer establishments with digital booking platforms because they provide faster transactions, accurate information, and convenient payment options. The researchers noted that customer satisfaction increases when reservation systems are easy to use, responsive, and reliable.

Johnson et al. also explained that automated systems contribute to better service quality because they reduce waiting time and provide instant confirmation to customers. The study emphasized that businesses implementing online reservation systems gain a competitive advantage by improving customer trust and loyalty. Moreover, digital reservation systems provide managers with useful reports and analytics that help monitor customer trends, booking frequency, and peak reservation periods.

In another foreign study, Gaur (2024) examined the role and effectiveness of online reservation systems in hotel operations and marketing strategies. The study found that reservation systems not only improve operational efficiency but also serve as an important marketing tool. According to Gaur, businesses with effective online reservation platforms attract more customers because they create a professional and modern image. The study also highlighted that online booking systems help businesses improve communication with customers through automated notifications, reminders, and promotional offers.

Gaur further explained that online reservation systems enhance organizational productivity by integrating reservation management, customer databases, and payment processing into a single platform. This integration reduces paperwork and improves coordination among staff members. Additionally, the study emphasized that businesses using digital systems can respond more quickly to customer inquiries and reservation requests, which contributes to higher customer satisfaction and retention.

The findings of these foreign studies suggest that online reservation systems are essential tools in modern hospitality management. They improve efficiency, minimize errors, enhance customer satisfaction, and strengthen business competitiveness. These studies provide valuable support for the development of an online reservation system for Messiah Inland Resort.

2.2 Local Studies

In the Philippines, several researchers have examined the importance of digital systems and application-based technologies in improving business operations, particularly in the hospitality and tourism industry. Dela Cruz and Santos (2022) conducted a study on application-based systems used by hospitality businesses. Their findings showed that digital systems improve data accuracy, operational efficiency, and customer service quality.

The researchers explained that manual processes often lead to data redundancy, misplaced records, and delayed transactions. By implementing application-based systems, businesses can centralize information and automate repetitive tasks such as reservations, billing, and customer record management. Dela Cruz and Santos also noted that digital systems reduce the workload of employees because information can be accessed and updated instantly.

Furthermore, the study emphasized that customers are more satisfied when businesses use technology to provide efficient and organized services. Customers appreciate the convenience of online booking, digital confirmations, and faster response times. The researchers concluded that adopting modern technology is essential for businesses that want to remain competitive in the growing hospitality industry.

Reyes (2021) also investigated the impact of digital tools on small businesses in the Philippines. The study revealed that digital systems help small enterprises manage daily operations more effectively while minimizing manual errors. According to Reyes, many small businesses still rely on traditional methods such as handwritten records and manual scheduling, which are time-consuming and inefficient.

The study found that introducing digital tools significantly improves productivity because information can be stored, retrieved, and managed more efficiently. Reyes emphasized that automated systems also enhance financial management by providing accurate transaction records and reports. Additionally, digital tools improve communication between businesses and customers by enabling online inquiries, reservations, and updates.

Reyes further explained that technology adoption among small businesses contributes to long-term growth and sustainability. Businesses that implement digital systems are better able to adapt to changing customer expectations and market demands. The findings support the idea that small resort businesses can greatly benefit from customized online reservation systems.

Another local study highlighted that information systems are becoming increasingly important in the tourism and hospitality industry because of the growing demand for fast and reliable services. Researchers observed that customers now expect businesses to provide convenient online platforms for reservations and inquiries. As a result, many businesses are transitioning from manual operations to computerized systems to improve service quality and operational efficiency.

The reviewed local studies demonstrate that digital systems play a vital role in improving business performance, customer satisfaction, and operational management.

These findings provide strong support for the proposed online reservation system for Messiah Inland Resort.

2.3 Related Literature

Literature on system development emphasizes the importance of proper planning, design, and implementation in creating reliable software applications. According to Pressman (2019), system analysis and design are essential processes in software engineering because they ensure that applications meet user needs and business requirements. Pressman explained that successful systems must be carefully planned to ensure functionality, efficiency, reliability, and usability.

The author further discussed that software development involves several stages, including planning, analysis, design, implementation, testing, and maintenance. Each stage contributes to the quality and effectiveness of the final system. Pressman highlighted that developers must understand the problems faced by users in order to create systems that provide practical and efficient solutions.

In relation to reservation systems, Alpaydin (2020) emphasized the importance of real-time availability, automated processes, and accurate record-keeping in improving service quality. According to the study, reservation systems must be designed to provide updated information instantly so customers can make informed booking decisions. Real-time functionality helps prevent scheduling conflicts and ensures efficient reservation management.

Alpaydin also explained that automation reduces human intervention, thereby minimizing errors and improving operational efficiency. Automated systems can handle multiple reservations simultaneously while maintaining accurate records. Additionally, the literature highlighted that reservation systems improve communication between businesses and customers through instant notifications, booking confirmations, and reminders.

Other literature suggests that digital transformation is becoming increasingly important in the hospitality industry because customers prefer convenient and technology-driven services. Businesses that implement online systems are more capable of meeting customer expectations and adapting to technological advancements. Researchers also noted that centralized data management enhances organizational performance by improving data accessibility, security, and reporting capabilities.

Moreover, literature related to hospitality management emphasizes that customer satisfaction is closely connected to service speed, convenience, and reliability. Online reservation systems address these factors by providing accessible and user-friendly platforms for customers. The integration of technology into business operations also contributes to improved decision-making because managers can easily access reports, customer information, and reservation histories.

The reviewed literature establishes that effective system development and implementation are necessary for improving operational processes and service quality. These concepts are relevant to the development of an online reservation system for Messiah Inland Resort.

2.4 Synthesis of the Reviewed Literature and Studies

The reviewed foreign and local studies consistently show that online reservation systems improve operational efficiency, data accuracy, customer satisfaction, and service quality. Foreign studies emphasized the role of automation and real-time updates in reducing booking conflicts, minimizing errors, and improving customer experiences. These studies also highlighted that online reservation systems strengthen business competitiveness by providing convenient and reliable services.

Local studies demonstrated that digital systems help businesses improve productivity, streamline operations, and reduce manual work. Researchers found that application-based systems are especially beneficial for small businesses because they simplify data management and enhance customer communication. The studies also showed that adopting digital technology contributes to business growth and sustainability.

Related literature emphasized the importance of proper system design and development in creating efficient and reliable applications. Authors discussed the significance of real-time functionality, centralized data management, and automation in improving reservation processes and service quality.

Despite the numerous studies on online reservation systems, limited research specifically focuses on small resort operations such as Messiah Inland Resort. Many existing studies concentrate on hotels and large hospitality establishments, leaving a gap in understanding the specific needs and challenges faced by smaller resorts. This study aims to address that gap by developing a customized online reservation system tailored to the operational requirements of Messiah Inland Resort.

The proposed system will provide real-time booking, centralized data management, and automated reservation processes to improve operational efficiency and customer satisfaction. Through this system, the resort can reduce manual errors, organize records effectively, and provide faster and more convenient services to customers. The reviewed literature and studies strongly support the relevance and significance of this research.

LITERATURE REVIEW

Overview of Online Reservation Systems

Online reservation systems are digital platforms designed to allow customers to reserve rooms, facilities, services, or accommodations through the internet. These systems are commonly integrated into websites or mobile applications that provide users with convenient access to booking services anytime and anywhere. Online reservation systems help businesses automate reservation processes, manage customer information efficiently, and provide real-time updates regarding availability and schedules.

According to Turban, King, Lee, and Chung (2000), internet-based systems have become an essential component of modern business operations, particularly in industries that rely heavily on customer service, such as hospitality and tourism. The researchers explained that electronic commerce and online systems improve communication, simplify transactions, and enhance organizational efficiency. Through internet technologies, businesses can provide faster and more convenient services while reducing the limitations associated with traditional manual processes.

The hospitality industry has greatly benefited from the development of online reservation systems because they simplify booking procedures for both businesses and customers. Traditionally, reservations were handled manually through phone calls, handwritten logs, or face-to-face transactions. These methods often resulted in delays, double bookings, misplaced records, and communication problems. With the introduction of computerized reservation systems, businesses can now manage bookings more accurately and efficiently.

Online reservation systems typically include features such as real-time availability checking, automated confirmation messages, customer account management, online payment options, and booking history records. These features help improve operational efficiency and customer satisfaction. Customers can easily view available schedules, select preferred services, and confirm reservations without the need to visit the establishment physically.

According to Laudon and Laudon (2018), information systems play a significant role in improving business operations by automating processes and organizing information effectively. The authors emphasized that computerized systems help businesses reduce operational costs, improve accuracy, and support better decision-making. In the hospitality industry, reservation systems contribute to smoother workflows because they centralize customer and booking information into a single database.

Furthermore, online reservation systems improve customer convenience and accessibility. Customers are able to make reservations at any time using computers or mobile devices connected to the internet. This accessibility is especially important in today’s fast-paced digital environment, where customers prefer quick and efficient services. Studies show that customers are more likely to choose businesses that offer online reservation options because these systems save time and provide immediate confirmation.

Another important advantage of online reservation systems is the reduction of human errors. Manual reservation methods are prone to mistakes such as incorrect scheduling, incomplete records, and lost information. Automated systems minimize these issues by storing data electronically and updating records instantly. According to Stair and Reynolds (2019), automation improves data accuracy and operational reliability because systems process information consistently and efficiently.

Online reservation systems also support business growth and competitiveness. Businesses that adopt digital technologies are better able to meet customer expectations and adapt to changing market demands. In the hospitality industry, modern reservation systems create a professional image and attract more customers because they provide organized and efficient services. Managers can also use reservation data and reports to analyze customer behavior, identify peak booking periods, and improve business strategies.

In addition, online reservation systems enhance communication between businesses and customers. Automated notifications, confirmation emails, and reminder messages help customers stay informed about their reservations. These communication features reduce misunderstandings and improve customer trust. Businesses can also respond more effectively to inquiries and concerns through integrated customer support functions.

Security and data management are also important aspects of online reservation systems. Modern systems use databases to securely store customer records, booking histories, and transaction details. Proper data management ensures that information can be retrieved easily when needed while maintaining confidentiality and accuracy. Effective database systems also support reporting and monitoring functions that help managers evaluate business performance.

The continuous advancement of technology has further improved online reservation systems by integrating mobile accessibility, cloud computing, and real-time synchronization. Mobile-friendly systems allow customers to make reservations through smartphones and tablets, increasing convenience and accessibility. Cloud-based systems enable businesses to access reservation data remotely, improving flexibility and operational management.

Despite the many benefits of online reservation systems, some businesses, particularly small establishments, still rely on manual methods due to limited resources or lack of technological knowledge. However, studies indicate that transitioning to computerized systems provides long-term advantages, including improved efficiency, better customer service, and increased profitability. As customer demand for digital services continues to grow, businesses are encouraged to adopt modern reservation technologies to remain competitive.

Overall, the literature suggests that online reservation systems are essential tools in the hospitality and tourism industry. They improve operational efficiency, enhance customer satisfaction, reduce errors, and support better management practices. These systems provide businesses with the ability to deliver faster, more reliable, and more convenient services to customers. The concepts and findings discussed in this literature are relevant to the development of an online reservation system for Messiah Inland Resort, as the proposed system aims to improve reservation management, customer service, and operational efficiency.

Systematic Review

A systematic review of online reservation systems reveals that digital platforms play a major role in improving business operations, particularly in the hospitality and tourism industries. Researchers consistently emphasize that automated reservation systems enhance efficiency, improve data accuracy, reduce operational errors, and increase customer satisfaction. These systems have become essential tools for businesses that aim to provide faster, more reliable, and more convenient services to customers.

Traditional reservation methods, such as manual booking through logbooks, phone calls, and paper records, are often associated with several problems. These include double bookings, lost reservation forms, delayed confirmations, inaccurate customer information, and inefficient scheduling. Because of these limitations, many businesses have shifted toward computerized reservation systems to improve operational management and customer service.

Studies show that online reservation systems automate booking processes and reduce the need for manual transactions. Automation minimizes human intervention, which significantly decreases the possibility of errors. Real-time updates allow customers and administrators to monitor room or facility availability instantly, helping businesses avoid scheduling conflicts and reservation overlaps. Researchers also note that automated systems improve communication by sending instant confirmations, reminders, and notifications to customers.

According to Turban and King (2000), the consumer purchasing process in electronic systems involves several important stages that are also applicable to online reservation systems. These stages demonstrate how customers interact with digital platforms to complete transactions efficiently and conveniently.

The first stage is identifying the need for a service or product. In the context of reservation systems, customers recognize the need for accommodations, resort bookings, event reservations, or other hospitality services. This stage motivates customers to search for available options that meet their preferences and requirements.

The second stage involves searching for available options. Through online reservation systems, customers can access information about available rooms, cottages, facilities, schedules, and services. Customers can browse websites or mobile applications to explore different choices without physically visiting the establishment. This accessibility provides convenience and saves time for both customers and businesses.

The third stage is comparing alternatives based on price, features, and availability. Online reservation systems allow customers to evaluate different options according to their budget, preferences, and desired services. Customers can compare room rates, amenities, schedules, and package inclusions before making decisions. This feature improves transparency and helps customers make informed choices.

The fourth stage is making a reservation or order. After selecting their preferred option, customers proceed with the booking process by entering personal information, choosing schedules, and confirming reservations. Automated systems simplify this process by providing user-friendly interfaces and organized booking procedures.

The fifth stage is processing payment or confirmation. Many online reservation systems provide digital payment methods such as online banking, electronic wallets, or credit card transactions. Some systems also allow reservation confirmation without immediate payment, depending on the business policy. Automated confirmation systems provide customers with instant verification that their reservations have been successfully processed.

The sixth stage is receiving confirmation and service details. Customers receive confirmation messages, booking references, receipts, and reservation schedules through email, text messages, or application notifications. These confirmations help customers verify their transactions and avoid misunderstandings regarding booking details.

The final stage involves providing feedback or requesting support if needed. Online systems often include customer support features, inquiry forms, ratings, and feedback mechanisms that allow customers to communicate concerns or share experiences. Businesses use this information to improve service quality and customer satisfaction.

These stages illustrate how online reservation systems simplify and organize the reservation process. The integration of digital technologies improves user experience by making transactions more efficient, accessible, and convenient. Customers are able to complete reservations quickly while businesses can manage operations more effectively.

Furthermore, systematic reviews indicate that online reservation systems contribute to better record management and data organization. Reservation details, customer information, and transaction histories are stored electronically in centralized databases, making retrieval and monitoring easier. Managers can also generate reports that help analyze booking trends, customer behavior, and business performance.

Researchers also highlight the importance of online reservation systems in improving customer satisfaction. Customers prefer businesses that provide accessible and convenient digital services because these systems reduce waiting time and provide immediate responses. Fast and reliable booking processes increase customer trust and loyalty, which are important factors in maintaining competitiveness in the hospitality industry.

Another important finding from systematic reviews is that online reservation systems improve business productivity. Employees spend less time performing repetitive administrative tasks because many processes are automated. This allows staff members to focus more on customer service and operational improvement. Businesses also benefit from reduced paperwork, improved coordination, and more accurate reporting systems.

In addition, online reservation systems support business growth by expanding market reach. Through internet-based platforms, businesses can attract customers from different locations and provide services beyond traditional face-to-face transactions. Mobile accessibility further enhances convenience because customers can make reservations using smartphones and tablets anytime and anywhere.

Despite the many advantages, some studies mention challenges related to system implementation, such as technical maintenance, internet dependency, and staff training requirements. Small businesses may face difficulties in adopting digital systems due to financial limitations or lack of technological expertise. However, researchers conclude that the long-term benefits of online reservation systems outweigh these challenges because of their positive impact on operational efficiency and customer satisfaction.

Overall, the systematic review demonstrates that online reservation systems are effective tools for improving reservation management, customer service, and business operations. The reviewed studies strongly support the development of a customized online reservation system for Messiah Inland Resort. By implementing real-time booking, centralized data management, and automated reservation processes, the proposed system can help improve efficiency, reduce manual errors, and provide better services to customers.

Direct vs. Indirect Systems in Online Platforms

Online platforms in the hospitality and tourism industry can generally be categorized into direct systems and indirect systems. These systems differ in the way customers interact with service providers and how reservations are processed. Both approaches are widely used in modern business operations and offer different advantages and challenges depending on the needs and goals of the business.

Direct systems involve transactions where customers interact directly with the service provider through the company's own website, mobile application, or reservation platform without the involvement of intermediaries. In this setup, resorts, hotels, and other hospitality businesses manage their own booking systems independently. Customers can directly access information about room availability, pricing, facilities, schedules, and services through the official platform of the business.

One major advantage of direct systems is that businesses maintain full control over reservations, pricing, customer information, and communication processes. This control allows businesses to manage their operations more effectively and provide personalized services to customers. Since there are no third-party intermediaries involved, businesses can avoid additional commission fees and maximize their revenue.

Direct reservation systems also improve data management because customer information is stored directly in the company's database. Businesses can easily monitor booking histories, customer preferences, and transaction records. This centralized management improves operational efficiency and supports better decision-making. Managers can analyze customer behavior and reservation trends to improve marketing strategies and customer services.

Another benefit of direct systems is improved communication between businesses and customers. Customers receive reservation confirmations, reminders, and updates directly from the establishment. This direct interaction helps build stronger customer relationships and increases customer trust and loyalty. Businesses can also respond more quickly to inquiries, special requests, and concerns.

According to Laudon and Laudon (2018), direct online systems help organizations improve efficiency by automating transactions and reducing operational costs. Automated reservation systems reduce manual work, minimize booking errors, and provide real-time updates that help businesses manage availability effectively. Customers also benefit from the convenience of making reservations anytime and anywhere using internet-connected devices.

Furthermore, direct systems contribute to brand identity and professionalism. Businesses with their own reservation platforms appear more organized and technologically advanced. Customers often perceive establishments with efficient online booking systems as more reliable and trustworthy. This can help attract more customers and strengthen the business's competitive advantage.

However, direct systems also require businesses to invest in technology, system maintenance, and technical support. Small businesses may face challenges in developing and maintaining their own reservation systems due to financial limitations or lack of technical expertise. Businesses must also ensure system security to protect customer data and maintain reliable online services.

On the other hand, indirect systems involve third-party platforms that act as intermediaries between customers and service providers. Examples of indirect systems include online travel agencies (OTAs), booking websites, and travel management platforms. Customers use these third-party platforms to search, compare, and reserve accommodations or services from different businesses in one centralized system.

Indirect systems provide several advantages, particularly in terms of market visibility and customer reach. Businesses listed on popular booking platforms can attract a larger number of customers because these websites are widely used and trusted by travelers. Third-party platforms often invest heavily in digital marketing and advertising, helping businesses gain exposure without needing extensive promotional efforts of their own.

Another advantage of indirect systems is convenience for customers. Online travel agencies allow users to compare prices, services, reviews, and availability from multiple establishments in one platform. This accessibility simplifies the decision-making process and improves the customer experience. Customers can easily identify the best options based on their preferences and budget.

Indirect systems are especially beneficial for small and newly established businesses that may not yet have strong online visibility. By partnering with third-party booking platforms, businesses can reach customers from different locations and increase reservation opportunities. This expanded market reach can contribute to business growth and profitability.

Despite these advantages, indirect systems also have disadvantages. One major limitation is the payment of commission fees to third-party platforms. Online travel agencies typically charge businesses a percentage of each reservation made through their system. These additional costs can reduce overall profits, especially for small businesses with limited budgets.

Another disadvantage is reduced control over customer data and communication. In many cases, third-party platforms manage customer interactions and reservation information. Businesses may have limited access to customer details, making it more difficult to build direct relationships and implement personalized marketing strategies. Dependence on intermediaries may also reduce a business's ability to establish strong brand recognition.

Additionally, businesses using indirect systems may face intense competition because customers can easily compare multiple establishments on the same platform. Price competition may pressure businesses to lower rates in order to attract customers, potentially affecting profitability.

According to Turban, Lee, King, and Chung (2000), businesses must carefully evaluate whether direct or indirect systems best suit their operational needs, financial resources, and marketing objectives. The researchers explained that direct systems are often more beneficial for small businesses because they provide better control over operations and reduce dependency on intermediaries. Direct systems also allow businesses to maintain ownership of customer data and avoid additional commission expenses.

However, Turban et al. also acknowledged that indirect systems can help businesses increase market exposure and attract more customers, particularly in highly competitive industries. Businesses may choose to combine both direct and indirect approaches in order to maximize visibility while maintaining operational control.

In the hospitality industry, the choice between direct and indirect systems depends on factors such as business size, budget, technological capability, target market, and long-term business goals. Large hotels and resorts often use both systems simultaneously, while smaller establishments may prioritize direct systems to reduce costs and strengthen customer relationships.

For small resorts such as Messiah Inland Resort, implementing a direct online reservation system can provide significant advantages in terms of operational control, customer management, and cost efficiency. A customized direct reservation system allows the resort to manage bookings independently, organize customer records effectively, and provide real-time reservation services without relying heavily on third-party platforms.

Overall, both direct and indirect systems play important roles in online business operations. Direct systems offer greater control, lower operational costs, and stronger customer relationships, while indirect systems provide increased visibility and broader market access. The reviewed literature suggests that businesses should carefully select the most appropriate system based on their specific operational requirements and business objectives.

Case Studies

Several case studies conducted in the hospitality and tourism industry highlight the effectiveness of online reservation systems in improving business performance, operational efficiency, and customer satisfaction. These studies demonstrate how digital reservation platforms help businesses overcome the limitations of traditional manual booking methods while providing more organized and reliable services.

Many small and medium-sized hospitality businesses that implemented online booking systems reported significant improvements in daily operations. Businesses observed reductions in booking errors, faster transaction processing, and more efficient management of customer reservations. Traditional manual reservation methods often resulted in scheduling conflicts, misplaced records, delayed confirmations, and inaccurate information. Through the use of automated reservation systems, these problems were minimized because booking information could be updated and monitored in real time.

One case study involving a small resort business revealed that implementing an online reservation system greatly improved operational efficiency during peak seasons. Before adopting the system, the resort relied on handwritten reservation logs and phone call inquiries, which frequently caused double bookings and delayed responses to customers. Employees struggled to manage the increasing number of reservations, especially during holidays and vacation periods.

After implementing a digital reservation platform, the business experienced major improvements in reservation management. Customers were able to book rooms and facilities online without the need for face-to-face transactions or phone calls. The system automatically updated room availability and generated instant confirmations for successful reservations. As a result, the resort reduced booking conflicts and improved customer satisfaction.

The case study also showed that employees became more productive because repetitive administrative tasks were automated. Staff members spent less time organizing reservation records manually and were able to focus more on customer service and operational management. Managers could also monitor reservations more effectively because all booking information was stored in a centralized database.

Another case study involving a medium-sized hotel demonstrated how online reservation systems improve customer convenience and service quality. The hotel introduced an internet-based booking platform that allowed customers to check room availability, compare prices, and make reservations through a mobile-friendly website. Customers appreciated the accessibility and convenience of the system because they could complete transactions anytime and anywhere.

The hotel management reported that customer satisfaction increased after the implementation of the online reservation system. Guests received immediate confirmation messages and detailed booking information, reducing confusion and misunderstandings. The system also provided automated reminders and notifications that helped customers stay informed about their reservations.

In addition, the case study emphasized the importance of real-time updates and centralized data storage in improving operational decision-making. Managers used reservation reports and booking data to identify customer trends, peak reservation periods, and frequently requested services. This information helped management improve staffing schedules, allocate resources more effectively, and create better marketing strategies.

Another important finding from case studies is that online reservation systems improve business competitiveness. Hospitality businesses that adopted digital platforms were perceived as more professional, organized, and technologically advanced. Customers preferred establishments that offered convenient online booking options because these systems reduced waiting time and simplified reservation procedures.

Case studies also revealed that businesses using online reservation systems were more capable of handling high customer demand during busy seasons. Automated systems could process multiple reservations simultaneously while maintaining accurate records and availability updates. In contrast, businesses relying on manual systems often experienced delays, confusion, and operational difficulties during peak periods.

Furthermore, digital reservation systems improved communication between businesses and customers. Automated confirmation emails, text notifications, and inquiry management tools allowed businesses to respond more efficiently to customer concerns and reservation requests. Effective communication contributed to stronger customer trust and increased customer loyalty.

Several studies also noted financial benefits associated with online reservation systems. Businesses reduced operational costs related to paperwork, manual record-keeping, and administrative labor. Automated systems streamlined reservation management processes, resulting in improved productivity and resource utilization. In the long term, these efficiencies contributed to increased profitability and business sustainability.

Despite the advantages, some case studies identified challenges in implementing online reservation systems. Small businesses sometimes faced difficulties related to technical maintenance, internet connectivity, and employee training. Initial development costs and system setup also required financial investment. However, researchers concluded that the long-term operational and customer service benefits outweighed these challenges.

According to Turban, King, Lee, and Chung (2000), electronic commerce systems provide businesses with opportunities to improve efficiency, communication, and customer interaction through internet-based technologies. The authors emphasized that digital systems support business growth by automating processes, organizing information, and improving accessibility for customers.

The reviewed case studies strongly support the implementation of an online reservation system for Messiah Inland Resort. The findings indicate that digital reservation platforms can improve booking management, reduce operational errors, enhance customer satisfaction, and strengthen business performance. By implementing a customized online reservation system with real-time booking and centralized data management, the resort can provide more efficient and reliable services to customers while improving overall operational efficiency.

Reference

Turban, E., King, D., Lee, J., & Chung, H. M. (2000). *Electronic Commerce: A Managerial Perspective*. Prentice Hall. ISBN 981-4058-34-3.

CHAPTER III

METHODOLOGY

3.1 Research Design

The study will utilize a developmental research design focusing on the design, development, implementation, and evaluation of the proposed Online Reservation and Management System for Messiah Inland Resort. This type of research design is appropriate because the study involves creating a software application intended to address existing operational problems and improve the efficiency of reservation management within the resort.

Developmental research is commonly applied in information technology and system development studies because it emphasizes the creation and enhancement of products, systems, or processes designed to solve identified problems. According to research methodologies in software development, developmental research involves systematic procedures in designing, developing, testing, and evaluating systems to ensure that they effectively meet user requirements and organizational objectives.

In this study, developmental research will guide the researchers throughout the entire process of system creation. The study aims to develop an efficient and user-friendly online reservation and management system that will improve the resort’s reservation procedures, customer service, and operational management. The researchers will follow a structured development process to ensure that the system functions properly and satisfies the needs of both administrators and customers.

The first stage of the research involves identifying the existing problems in the current manual reservation system of Messiah Inland Resort. Through observation, interviews, and data gathering, the researchers will analyze the challenges encountered by the resort management, such as booking conflicts, delayed transactions, inaccurate records, and difficulties in managing customer information. Understanding these issues is important in determining the necessary features and functions of the proposed system.

The second stage involves gathering system requirements. The researchers will collect information regarding the needs and expectations of the resort staff, administrators, and potential users. This process includes identifying the necessary functionalities of the system such as online booking, reservation management, customer record storage, availability monitoring, payment tracking, and report generation. Requirement gathering ensures that the system is aligned with the operational needs of the resort.

The third stage is system planning and design. During this phase, the researchers will create the overall structure and framework of the proposed application. This includes designing the system architecture, user interface, navigation flow, and database structure. The interface design will focus on user-friendliness, accessibility, and simplicity to ensure that users can easily navigate and operate the system. The database design will organize reservation records, customer information, transaction details, and system reports in a centralized and secure manner.

The fourth stage is system development or coding. In this stage, the researchers will convert the planned design into a functional application using appropriate programming languages, software tools, and database management systems. The system will include features such as online reservation forms, real-time availability updates, customer account management, automated confirmations, and administrative controls. The researchers will ensure that the system performs efficiently and accurately according to the identified requirements.

The fifth stage involves system testing and debugging. The developed system will undergo several testing procedures to identify errors, technical issues, and functionality problems. Testing is important to ensure that all system features operate correctly and reliably. Different types of testing may be conducted, including functionality testing, usability testing, compatibility testing, and performance testing. Errors identified during testing will be corrected and improved to enhance the overall quality of the system.

After the development and testing phases, the proposed system will be evaluated based on specific quality criteria such as functionality, efficiency, reliability, usability, and performance. The evaluation process will determine whether the system effectively addresses the problems identified in the existing manual reservation process. The researchers will assess if the system improves booking management, reduces operational errors, enhances data accuracy, and provides faster services to customers.

Feedback from intended users such as resort staff, administrators, and selected customers will also be gathered during the evaluation process. Their feedback and suggestions will help identify areas that may require improvement or additional features. User evaluation is important because it measures the acceptability and practicality of the system in real-world operations.

Furthermore, the developmental research design supports continuous improvement of the system. Based on the evaluation results and user feedback, the researchers may revise or enhance certain system features to improve performance and usability. This process ensures that the final system meets the standards and expectations of the users and contributes to the improvement of the resort’s operational management.

The use of developmental research design is highly suitable for this study because it provides a systematic and organized approach to software creation and evaluation. Through this research design, the researchers will be able to develop an effective Online Reservation and Management System that can help Messiah Inland Resort improve operational efficiency, customer satisfaction, and reservation management processes.

Through this research design, the study aims to produce a functional and reliable prototype that can improve reservation management and support the daily operations of Messiah Inland Resort.

3.2 System Development Methodology

The study will use the Agile Development Model as the system development methodology for the proposed Online Reservation and Management System for Messiah Inland Resort. Agile methodology is a widely used software development approach that emphasizes flexibility, continuous improvement, collaboration, and iterative development. This methodology is suitable for the study because it allows the researchers to develop the system in phases while continuously evaluating and improving its features based on user feedback and system requirements.

The Agile Development Model focuses on developing software incrementally through repeated cycles known as iterations or sprints. Instead of completing the entire system at once, the development team creates smaller functional components that are continuously tested and improved. This approach helps ensure that the system remains aligned with user needs and minimizes the risk of major development errors.

Agile methodology also promotes close communication between developers and users throughout the development process. Through regular evaluation and feedback, the researchers can identify problems early and make necessary modifications to improve the system’s functionality, usability, and performance. The flexibility of Agile development makes it highly effective for system projects that may require adjustments and enhancements during the development process.

The development of the proposed system will follow the following phases of the Agile Development Model:

1. Requirements Analysis

The first phase of the methodology is Requirements Analysis. In this phase, the researchers will gather information regarding the needs, problems, and expectations of the users and administrators of Messiah Inland Resort. Data gathering methods such as interviews, observations, and discussions will be conducted to identify the issues encountered in the current manual reservation system.

The researchers will analyze the existing reservation process to determine the necessary features and functionalities of the proposed system. These may include online booking, customer registration, reservation monitoring, room or facility availability checking, payment management, automated notifications, and report generation. The collected information will serve as the foundation for designing and developing the system.

Requirements analysis is important because it ensures that the system addresses the actual operational needs of the resort. Proper identification of user requirements helps avoid unnecessary features and ensures that the system will be useful, efficient, and user-friendly.

2. System Design

The second phase is System Design. In this phase, the researchers will create the overall structure and framework of the proposed Online Reservation and Management System. The design process includes developing system architecture, flowcharts, data flow diagrams, entity relationship diagrams, database structures, and user interface designs.

The researchers will design the system interface to ensure simplicity, accessibility, and ease of navigation for users. Input forms, reservation pages, dashboards, and administrative controls will be carefully planned to provide a smooth and organized user experience.

Database design will also be developed during this phase. The database will store customer information, reservation records, payment details, schedules, and reports in a centralized system. Proper database structure is essential for maintaining data accuracy, security, and efficient retrieval of information.

The system design phase serves as the blueprint for the development stage. A well-designed system structure helps developers implement features accurately and efficiently while minimizing technical issues during coding.

3. Development

The third phase is Development. During this phase, the researchers will begin coding and integrating the different components of the system based on the approved system design. Appropriate programming languages, frameworks, and database management tools will be used to build the application.

The researchers will develop essential features such as customer registration, online reservation forms, real-time booking updates, availability monitoring, payment tracking, administrative management, and report generation. The integration of these components will allow the system to function as a complete online reservation and management platform.

Agile development encourages incremental coding and continuous improvement. The researchers may develop the system in smaller modules or sections and test them regularly to ensure proper functionality. This approach allows problems to be identified and corrected early in the development process.

The development phase is critical because it transforms the planned system design into a functional and operational application. Proper coding practices and system integration are necessary to ensure system efficiency, reliability, and security.

4. Testing

The fourth phase is Testing. After the system components have been developed, the researchers will conduct various testing procedures to identify errors, bugs, and technical issues. Testing ensures that the system performs according to its intended functions and meets the requirements identified during the analysis phase.

Different testing methods may be conducted, including functionality testing, usability testing, compatibility testing, and performance testing. Functionality testing verifies whether system features operate correctly, while usability testing evaluates the ease of use and user satisfaction of the application.

The researchers will also test the accuracy of reservation processing, database storage, report generation, and real-time updates to ensure that the system provides reliable and efficient services. Any identified problems or errors will be corrected and improved during this phase.

Testing is essential because it helps ensure the quality, stability, and reliability of the proposed system before implementation. A properly tested system minimizes operational issues and improves user confidence and satisfaction.

5. Implementation and Evaluation

After successful testing, the system will proceed to implementation and evaluation. During implementation, the proposed Online Reservation and Management System will be introduced to the intended users of Messiah Inland Resort. Resort staff and administrators will be allowed to use the system in actual reservation and management operations.

The researchers will gather feedback from users regarding the functionality, efficiency, reliability, and usability of the system. User feedback is important because it helps determine whether the system effectively addresses the operational problems encountered in the previous manual process.

The evaluation results will also help identify areas for further improvement and enhancement. Since Agile methodology supports continuous development, additional modifications and updates may be implemented to further improve system performance and user experience.

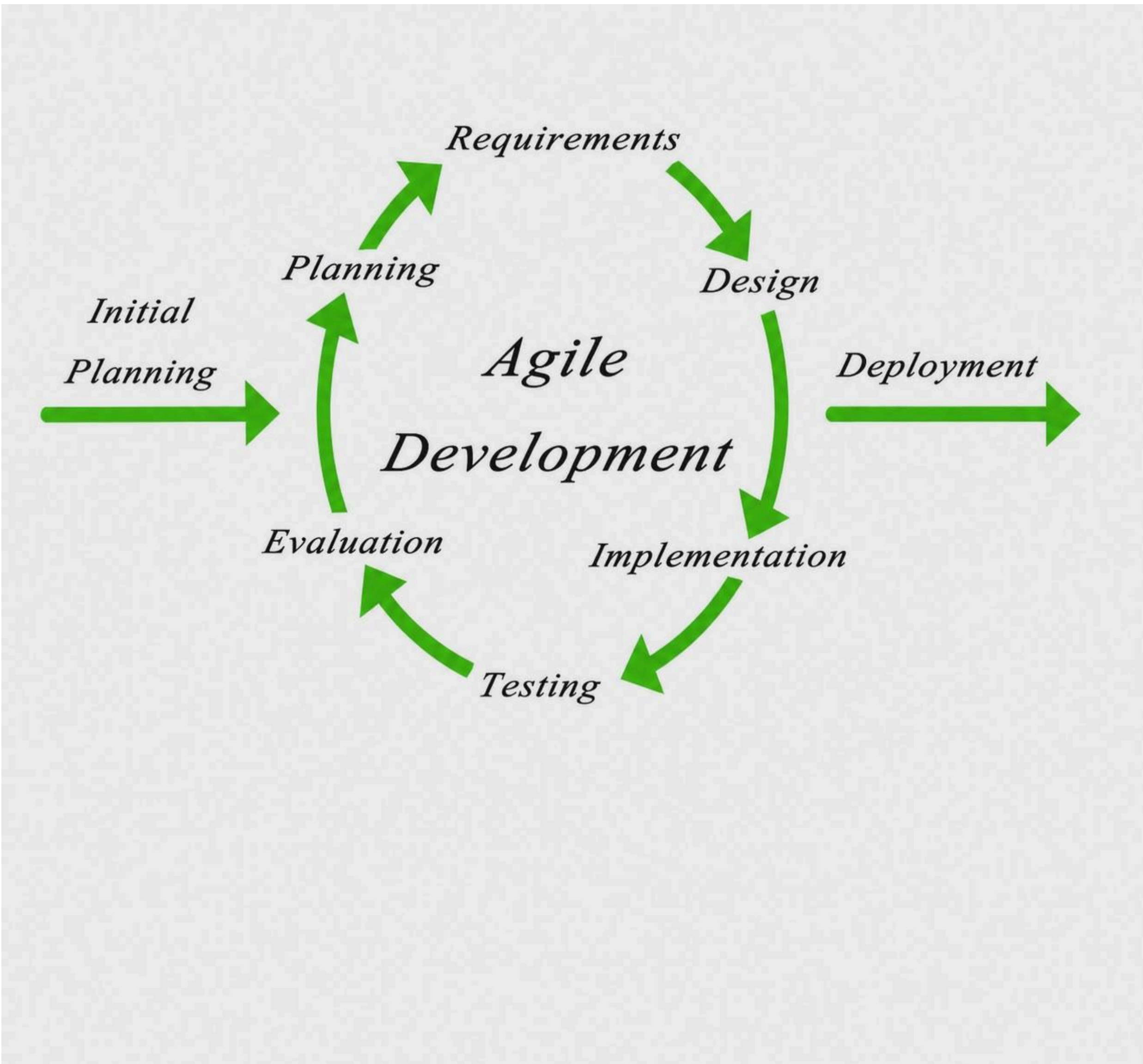
Overall, the Agile Development Model provides a flexible and organized approach to system development. Its iterative process allows continuous testing, evaluation, and improvement throughout the project. By using this methodology, the researchers can ensure that the proposed Online Reservation and Management System for Messiah Inland Resort is functional, reliable, efficient, and capable of meeting the operational needs of the resort.

The Agile Development Model is used in this study because it allows continuous improvement through iterative development and user feedback. This ensures that the system meets the needs of Messiah Inland Resort while allowing quick adjustments during the development process.

Additionally, Agile encourages collaboration and regular testing in each phase. This helps identify and fix errors early, improving the system's reliability, usability, and overall performance.

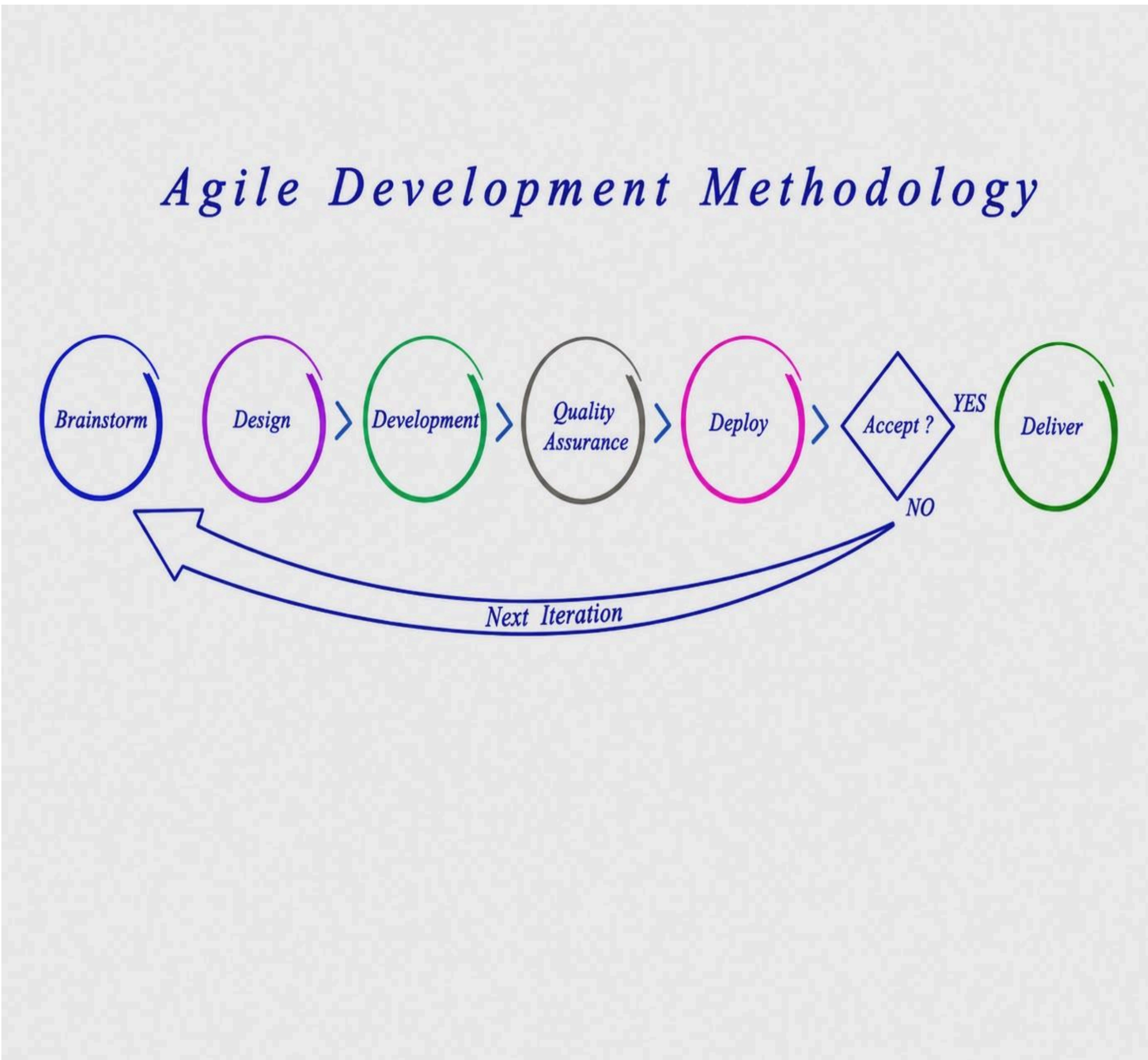
PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education



PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education



3.3 System Architecture

The proposed Online Reservation and Management System for Messiah Inland Resort will follow a client-server architecture. This type of architecture is widely used in modern web-based and mobile-based applications because it allows efficient communication between users and centralized system resources. In a client-server setup, the client side is responsible for user interaction, while the server side manages data processing, database operations, and system functionalities.

The client-server architecture is appropriate for the proposed system because it supports centralized management of reservation data, real-time updates, secure information storage, and multi-user accessibility. Through this architecture, customers and administrators can access the system using internet-connected devices such as computers, laptops, tablets, or smartphones.

In the proposed system, the client refers to the web or mobile interface used by customers, resort staff, and administrators. The client interface serves as the front-end component of the system where users interact with the application. Customers can use the interface to create accounts, check room or facility availability, submit reservations, view booking confirmations, and access reservation details. Administrators and resort staff can also use the interface to manage reservations, monitor bookings, update schedules, and generate reports.

The client interface is designed to be user-friendly, accessible, and responsive to ensure smooth interaction for users. A responsive design allows the system to adapt to different screen sizes and devices, improving accessibility and convenience for customers using mobile phones or tablets. The interface also provides navigation menus, forms, dashboards, and notification features that simplify the reservation process.

The server serves as the central processing unit of the system. It is responsible for handling requests from users, processing reservation transactions, validating information, and managing communication between the client interface and the database. When users submit reservation requests or access information through the client interface, the server receives and processes these requests before sending the appropriate responses back to the users.

The server also manages the database system, which stores all essential information related to reservations and resort operations. The database contains customer records, booking schedules, payment details, room availability, transaction histories, and system reports. Centralized database management ensures that information is organized, accurate, and accessible whenever needed.

One of the major advantages of centralized database management is real-time data synchronization. When a reservation is made or updated, the changes are automatically reflected throughout the system. This feature helps prevent double bookings, scheduling conflicts, and outdated information. Real-time updates also allow administrators and customers to access the most current reservation data.

The proposed system architecture also supports security and data protection. Since reservation and customer information are stored in a centralized server, security measures such as authentication, password protection, and controlled access can be implemented more effectively. Administrators can manage user permissions to ensure that only authorized personnel can access sensitive information.

Additionally, the server is responsible for processing machine learning predictions for matching. Machine learning functionality may be integrated into the system to assist in analyzing customer preferences, predicting booking patterns, or recommending suitable accommodations and services based on previous reservation data. The use of machine learning enhances system intelligence and improves decision-making processes within the application.

For example, the system may analyze customer booking behavior to identify frequently selected facilities, preferred schedules, or seasonal reservation trends. These predictions can help administrators optimize resource allocation, improve customer recommendations, and enhance service quality. Machine learning can also support automated matching processes by suggesting available rooms or facilities that best fit customer requirements and preferences.

The communication process within the client-server architecture follows a request-response model. When users perform actions such as logging in, checking availability, or submitting reservations, the client sends requests to the server through the internet. The server processes the requests, retrieves or updates information from the database, and sends responses back to the client interface. This interaction allows users to receive immediate updates and real-time feedback regarding their transactions.

Another important advantage of client-server architecture is scalability. As the number of users and reservations increases, the system can be expanded by upgrading server resources or improving network infrastructure. This flexibility ensures that the system can continue to perform efficiently even during peak reservation periods or increased operational demands.

The architecture also improves system maintenance and management because updates and modifications can be applied directly on the server side without requiring users to reinstall applications or modify their devices. Centralized maintenance simplifies troubleshooting, system upgrades, and security monitoring.

Furthermore, the client-server model supports reliable backup and recovery procedures. Reservation data and transaction records can be backed up regularly to prevent data loss caused by technical failures or system errors. This helps ensure business continuity and protects important operational information.

Overall, the client-server architecture provides an efficient, secure, and organized framework for the proposed Online Reservation and Management System for Messiah Inland Resort. Through the integration of web and mobile interfaces, centralized database management, real-time synchronization, and machine learning capabilities, the system can improve reservation processing, customer service, and operational efficiency. The architecture supports the system's goal of providing a modern, reliable, and user-friendly platform for managing resort reservations and operations.

3.4 Tools and Technologies Used

- Frontend: React JSX, Vite, CSS
- Backend: Node.js + Express.js
- Database: MySQL
- Development Tools: Visual Studio Code
- Laptop: Acer

3.5 Data Gathering Techniques

Data gathering is an important part of the study because it allows the researchers to collect accurate, reliable, and relevant information regarding the current reservation process of Messiah Inland Resort. Through proper data collection procedures, the researchers will be able to identify the problems encountered in the existing manual reservation system and determine the specific requirements needed for the development of the proposed Online Reservation and Management System.

The gathered information will serve as the foundation for system planning, design, development, and evaluation. The researchers will use several data gathering methods to ensure that all necessary information is collected comprehensively and accurately. The following methods will be used in this study:

Interviews

Structured interviews will be conducted with the resort management, staff, and other individuals directly involved in the reservation and operational processes of Messiah Inland Resort. The purpose of the interviews is to gather detailed information regarding the current reservation procedures, operational challenges, and expectations for the proposed system.

The researchers will prepare a set of interview questions focusing on topics such as reservation handling, booking confirmation procedures, customer record management, availability monitoring, payment processing, and report generation. Through these interviews, the researchers will gain a deeper understanding of the existing workflow and identify the specific problems experienced by the resort staff and administrators.

The interview process will also allow the respondents to express their suggestions, preferences, and expectations regarding the proposed online reservation system. Their insights will help the researchers determine the necessary features and functionalities that should be included in the system. Information gathered through interviews is important because it provides firsthand data directly from the intended users of the application.

Observation

Observation will also be used as a data gathering method in the study. The researchers will observe the current manual reservation process of Messiah Inland Resort to understand how operations are performed in actual practice. This includes observing how bookings are received, how customer information is recorded, how room or facility availability is checked, and how reservation records are managed.

Through observation, the researchers will identify inefficiencies, delays, repetitive tasks, and possible errors present in the existing system. Observation allows the researchers to examine the actual workflow and operational environment of the resort without relying solely on verbal explanations from respondents.

The observation process will also help determine the time consumed by manual reservation activities and identify areas where automation can improve efficiency and productivity. By carefully analyzing the current processes, the researchers can design a more effective and organized reservation system that addresses the weaknesses of the manual method.

Furthermore, observation will help the researchers understand how employees interact with reservation records and customers. This information is important in designing a user-friendly and practical system interface that matches the daily operational needs of the resort.

Document Review

The researchers will also conduct a review of existing documents and records used by Messiah Inland Resort in its current reservation process. These documents may include reservation logs, booking forms, customer records, payment receipts, reservation schedules, and transaction reports.

The purpose of document review is to examine how information is currently stored, organized, and managed within the resort. By reviewing these records, the researchers can identify the type of data required in the proposed system and determine how information flows within the reservation process.

Document review also helps the researchers identify issues related to data redundancy, incomplete records, inaccurate information, and difficulties in retrieving reservation details. Understanding the structure and content of existing documents is important for designing a well-organized database system that improves data storage, accessibility, and management.

Additionally, the review of documents will help ensure that the proposed system includes all essential information needed for reservation management and reporting. It will guide the researchers in developing appropriate database fields, forms, and system reports that align with the operational requirements of the resort.

Questionnaire

If applicable, questionnaires may also be distributed to resort staff, administrators, and selected customers to gather additional information regarding the current reservation system and user preferences for the proposed online platform. The questionnaire method allows the researchers to collect feedback from multiple respondents efficiently.

The questionnaires may contain questions related to user satisfaction, common reservation problems, preferred system features, ease of use, and expectations regarding online reservation services. Respondents may also provide suggestions on how the reservation process can be improved through automation and digital technology.

Questionnaires are useful in obtaining quantitative and qualitative data that support the findings gathered from interviews and observations. The responses collected can help the researchers evaluate the level of user acceptance and identify features that are most important to potential users of the system.

The use of questionnaires also allows the researchers to gather customer perspectives regarding reservation convenience, accessibility, and service expectations. This information is valuable in ensuring that the proposed system is customer-friendly and responsive to user needs.

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

Importance of Data Gathering Procedures

The use of multiple data gathering methods ensures that the study collects comprehensive and accurate information from different sources. Interviews provide detailed insights from staff and administrators, observation reveals actual operational practices, document review examines existing records and data structures, while questionnaires gather feedback from users and customers.

Combining these methods improves the reliability and validity of the gathered data because information can be verified and compared across different sources. The collected data will guide the researchers in identifying operational problems, defining system requirements, and designing appropriate solutions for the proposed Online Reservation and Management System.

Through these data gathering procedures, the researchers will be able to develop a system that effectively addresses the needs of Messiah Inland Resort. The information collected will support the creation of a functional, efficient, reliable, and user-friendly online reservation system that improves booking management, customer service, and overall operational efficiency.

SYSTEM DESIGN AND DEVELOPMENT

The system design and development phase focuses on creating a clear, organized, and efficient structure for the proposed Online Reservation and Management System for Messiah Inland Resort. This phase is important because it transforms the gathered requirements and identified problems into a functional software application that meets the needs of both customers and resort management.

System design and development involve planning the system structure, creating interfaces, organizing databases, developing program functionalities, and integrating system components. Proper system design ensures that the application will operate efficiently, securely, and reliably while providing a user-friendly experience for all users.

The researchers aim to develop a system that improves reservation management, minimizes manual errors, provides real-time booking updates, and enhances customer convenience. The following components are included in the system design and development phase:

User Interface (UI) Design

The User Interface (UI) Design focuses on creating a simple, intuitive, and user-friendly interface for customers, administrators, and resort staff. The interface serves as the visual component of the system where users interact with the application to perform reservation and management activities.

The researchers will design the interface to ensure that users can easily navigate the system and access its features without difficulty. The interface will include menus, forms, dashboards, buttons, and navigation tools that guide users throughout the reservation process.

For customers, the interface will allow them to create accounts, check room or facility availability, make reservations, view booking confirmations, and manage reservation details conveniently. The system will provide clear instructions and organized layouts to simplify the booking process.

For administrators and resort staff, the interface will include management tools for monitoring reservations, updating schedules, managing customer records, confirming bookings, and generating reports. Administrative dashboards will provide organized displays of reservation information and operational data.

The UI design will also focus on responsiveness and accessibility. The system will be designed to function properly on different devices such as desktop computers, laptops, tablets, and smartphones. Responsive design improves accessibility and allows users to access the system conveniently from various platforms.

A properly designed user interface improves user satisfaction because it reduces confusion, minimizes operational errors, and enhances the overall user experience.

Database Design

Database Design focuses on creating a well-structured and organized database that stores and manages important information related to the reservation system. The database serves as the central storage component of the application and plays an important role in maintaining data accuracy, consistency, and accessibility.

The proposed database will contain information such as customer profiles, reservation records, room or facility details, booking schedules, payment transactions, availability status, and report data. Each category of information will be organized into appropriate tables and relationships to ensure efficient data management.

The researchers will design the database to support accurate and secure data storage. Proper database structure helps prevent data redundancy, duplication, and inconsistencies. It also improves the speed and efficiency of retrieving information whenever needed.

Database security is also an important consideration in the design process. Access controls and authentication procedures may be implemented to protect sensitive customer and reservation data from unauthorized access.

The database system will support real-time updates, allowing reservation information to be automatically synchronized throughout the application. This feature helps prevent double bookings and ensures that customers and administrators always access updated availability information.

In addition, the database will support reporting and monitoring functions. Administrators will be able to generate reports related to reservations, transactions, customer records, and booking trends, which can assist in operational analysis and decision-making.

System Architecture

The system will be designed using a structured and modular architecture to ensure efficiency, reliability, scalability, and maintainability. Modular architecture divides the system into separate components or modules that perform specific functions. This approach improves organization and makes the system easier to manage, maintain, and upgrade in the future.

The proposed Online Reservation and Management System will follow a client-server architecture. Users will access the system through web or mobile interfaces, while the server handles data processing, database management, and system functionalities.

The architecture will support smooth communication between the user interface and the database system. When customers submit reservation requests, the server processes the information and updates the database in real time. This interaction ensures accurate reservation management and immediate system responses.

The modular structure also allows the future integration of additional features such as automated notifications, payment gateways, report generation, and customer feedback systems. Since the system is designed to be scalable, future enhancements can be implemented without significantly affecting existing functionalities.

System reliability and performance are also considered in the architectural design. The system will be developed to handle multiple users and reservation transactions efficiently, especially during peak booking periods. Proper architecture helps ensure system stability, faster processing, and improved operational performance.

Furthermore, the architecture supports easier maintenance and troubleshooting because system components are organized systematically. Developers can modify or improve individual modules without affecting the entire application.

Through proper system design and development, the researchers aim to create a functional, efficient, reliable, and user-friendly Online Reservation and Management System for Messiah Inland Resort. The proposed system is expected to improve reservation processing, reduce manual errors, enhance customer satisfaction, and support better operational management for the resort.

3.6 Testing Procedures

Testing procedures are an essential part of system development because they ensure that the proposed Online Reservation and Management System for Messiah Inland Resort functions correctly and meets all required specifications, objectives, and user expectations. System testing helps identify errors, improve system performance, ensure data accuracy, and evaluate the reliability and usability of the application before it is fully implemented.

The testing phase is important because even well-designed systems may contain technical issues, coding errors, or functionality problems that could affect system performance and user experience. Through proper testing procedures, the researchers can identify and correct these problems early to ensure that the system operates efficiently and reliably.

The primary purpose of testing is to verify that the developed system performs according to its intended functions. Testing also helps determine whether the system can process reservations accurately, manage customer information properly, provide real-time updates, and support smooth interactions between users and administrators.

In this study, several testing methods will be conducted to evaluate different aspects of the proposed system. These testing procedures include Unit Testing, Integration Testing, and System Testing.

Unit Testing

Unit Testing involves testing the individual components or modules of the system separately to ensure that each function performs correctly on its own. This type of testing focuses on examining the smallest functional parts of the system before integrating them into the complete application.

The researchers will test each feature independently to verify that it produces the expected results. Examples of components that will undergo unit testing include the login module, registration module, reservation form, availability checking function, payment processing feature, customer information management, and report generation functions.

For example, the login module will be tested to ensure that users can successfully log in using valid credentials while preventing unauthorized access through invalid login attempts. The reservation module will also be tested to verify that booking information is properly saved and updated in the database.

Unit testing is important because it helps identify coding errors, logic problems, and functionality issues within individual modules before they are combined into the complete system. Detecting errors at an early stage reduces development time and prevents problems from affecting other parts of the application.

Additionally, unit testing helps ensure that each feature meets the specified functional requirements and performs accurately under different conditions. By thoroughly testing individual components, the researchers can improve system reliability and reduce the risk of operational failures.

Integration Testing

Integration Testing is conducted after unit testing to ensure that the different modules and components of the system work together properly. While unit testing focuses on individual functions, integration testing examines how these functions interact with one another within the complete system.

The purpose of integration testing is to identify issues related to data communication, module interaction, and system coordination. The researchers will evaluate whether features such as reservation processing, user management, database operations, and reporting functions operate smoothly when combined together.

For example, when a customer submits a reservation, the system must correctly process the booking request, update room availability, store customer information in the database, and generate confirmation messages. Integration testing verifies that these related components communicate and function properly as a unified process.

The researchers will also test how administrative functions interact with customer modules. This includes verifying whether administrators can view reservation updates in real time, modify booking information, and retrieve accurate reports from the database.

Integration testing is important because even if individual modules function correctly during unit testing, problems may still occur when modules interact with each other. Common issues identified during integration testing include data transfer errors, synchronization problems, inconsistent outputs, and communication failures between system components.

By performing integration testing, the researchers can ensure that the entire system workflow operates efficiently and without interruption. This helps improve system stability, data consistency, and overall operational performance.

System Testing

System Testing involves evaluating the complete Online Reservation and Management System as a whole to ensure that it performs effectively under real-world conditions. This testing method focuses on the overall functionality, performance, reliability, security, and usability of the application.

The researchers will test the entire system to verify that all integrated modules work correctly according to the specified system requirements and objectives. System testing simulates actual usage scenarios to determine whether the application can handle reservation transactions, data processing, and user interactions efficiently.

During system testing, the researchers will evaluate different aspects of the application, including:

- Accuracy of reservation processing
- Real-time availability updates
- Database performance and data retrieval
- User authentication and access control
- System responsiveness and speed
- Ease of navigation and usability
- Error handling and system recovery
- Compatibility with different devices and browsers

The system will also be tested to determine how it performs during multiple simultaneous transactions, particularly during peak reservation periods. This helps ensure that the system remains stable and responsive even when handling high volumes of users and bookings.

Usability testing will also be included in system testing to evaluate the user experience of customers, resort staff, and administrators. Users may be asked to perform specific tasks such as creating reservations, updating booking details, or generating reports to determine whether the system is easy to use and understand.

Security testing may also be performed to ensure that customer information and reservation data are protected from unauthorized access or data loss. This includes testing user authentication, password protection, and access permissions.

System testing is essential because it verifies whether the application is ready for actual implementation and operation. Through this process, the researchers can identify remaining technical issues, performance limitations, or usability concerns that need improvement before deployment.

Importance of Testing Procedures

Testing procedures are necessary to ensure the quality, reliability, and effectiveness of the proposed Online Reservation and Management System. Proper testing helps prevent operational failures, improves system performance, and ensures that users can interact with the system efficiently and confidently.

By conducting unit testing, integration testing, and system testing, the researchers can thoroughly evaluate all aspects of the application and ensure that it satisfies the operational needs of Messiah Inland Resort. These testing methods help confirm that the system can accurately manage reservations, process customer information, provide real-time updates, and support efficient resort operations.

Furthermore, testing procedures contribute to user satisfaction because they help create a stable, user-friendly, and dependable system. A properly tested application minimizes errors, improves security, and enhances the overall quality of service provided to customers and resort staff.

Through these testing procedures, the researchers aim to ensure that the Online Reservation and Management System for Messiah Inland Resort is stable, accurate, efficient, and suitable for practical implementation and long-term use.

3.7 Evaluation Criteria

The proposed Online Reservation and Management System for Messiah Inland Resort will be evaluated using several important criteria to ensure that it meets the required standards, functions properly, and effectively serves its intended purpose. Evaluation is an essential part of system development because it determines whether the application successfully addresses the operational problems identified in the existing manual reservation process.

The evaluation process will assess the overall quality, effectiveness, usability, and reliability of the system. It will also determine whether the developed application satisfies the needs of customers, resort staff, and administrators. The results of the evaluation will help identify strengths, weaknesses, and possible areas for improvement before the full implementation of the system.

The following evaluation criteria will be used in assessing the Online Reservation and Management System for Messiah Inland Resort:

Functionality

Functionality refers to the ability of the system to perform all required features and operations correctly according to the specified requirements and objectives. This criterion evaluates whether the system can properly execute its intended functions without errors or failures.

The researchers will assess whether important system features such as online reservation handling, customer registration, user authentication, booking confirmation, payment recording, reservation updates, report generation, and real-time availability monitoring are functioning accurately and efficiently.

The functionality evaluation also examines whether the system produces correct outputs based on user inputs and whether all modules interact properly with one another. A functional system should support smooth reservation processing and provide accurate information to users and administrators.

Ensuring functionality is important because the effectiveness of the system largely depends on its ability to perform the tasks it was designed to accomplish.

Usability

Usability measures how easy, convenient, and user-friendly the system is for customers, resort staff, and administrators. This criterion focuses on the overall user experience and evaluates whether users can interact with the system efficiently and comfortably.

The evaluation of usability includes examining the clarity of the interface design, ease of navigation, accessibility of features, readability of content, and simplicity of the reservation process. Users should be able to complete tasks such as creating reservations, checking availability, and updating booking information without confusion or difficulty.

The researchers will also assess whether the system interface is responsive and compatible with different devices such as desktop computers, laptops, tablets, and smartphones. A well-designed and user-friendly interface improves user satisfaction and reduces operational errors caused by misunderstandings or navigation difficulties.

Usability is important because even a technically functional system may fail if users find it difficult to operate. A user-friendly system encourages acceptance and efficient use among customers and resort personnel.

Security

Security refers to the protection of customer information, reservation records, and system data against unauthorized access, misuse, loss, or cyber threats. Since the system handles sensitive information such as customer details and booking transactions, implementing proper security measures is essential.

The evaluation of security includes examining user authentication processes, password protection, access controls, and data privacy mechanisms. The researchers will ensure that only authorized users can access administrative functions and confidential information.

The system will also be evaluated for its ability to protect stored data from accidental loss or unauthorized modification. Proper security measures help maintain customer trust and ensure the confidentiality and integrity of reservation records.

Security is a critical aspect of system evaluation because weak security can result in data breaches, operational disruptions, and loss of customer confidence.

Efficiency

Efficiency measures how quickly and effectively the system performs tasks and processes information. This criterion evaluates the system’s ability to complete reservation-related operations with minimal delays and resource consumption.

The researchers will assess how efficiently the system handles activities such as reservation processing, database updates, customer registration, information retrieval, and report generation. The system should be capable of processing requests quickly and accurately even during periods of high reservation activity.

Efficiency also involves evaluating whether the system minimizes repetitive manual work and improves operational productivity. An efficient reservation system helps reduce waiting time, improve workflow management, and enhance customer service.

The evaluation of efficiency is important because users expect fast and responsive digital services, especially in online reservation platforms.

Reliability

Reliability refers to the system's ability to operate consistently and continuously without frequent errors, crashes, or failures. A reliable system should maintain stable performance under normal operating conditions and provide dependable services to users.

The researchers will evaluate whether the system can consistently process reservations, store data accurately, and maintain stable functionality over extended periods of use. The system should also recover properly from errors or interruptions without causing data loss or operational problems.

Reliability testing may include evaluating system stability during multiple simultaneous transactions and examining how the application performs during peak booking periods.

A reliable system is important because operational interruptions and technical failures can negatively affect customer satisfaction and resort operations.

Accuracy

Accuracy evaluates whether the system provides correct, updated, and error-free information to users and administrators. This criterion is especially important in reservation systems because inaccurate information can lead to booking conflicts, customer dissatisfaction, and operational problems.

The researchers will assess whether the system accurately records customer details, reservation schedules, payment information, and room or facility availability. Real-time updates should properly reflect reservation changes and current availability status.

The system should also generate accurate reports and transaction records that can assist administrators in monitoring operations and making decisions.

Ensuring system accuracy is essential because incorrect reservation data may result in scheduling conflicts, double bookings, and financial inconsistencies.

Maintainability

Maintainability refers to how easy it is to update, modify, repair, or improve the system when necessary. Since technology and operational requirements may change over time, the system should be designed in a way that supports future maintenance and enhancement.

The researchers will evaluate whether the system architecture and source code are organized, modular, and easy to manage. Proper documentation, structured coding practices, and modular design contribute to easier maintenance and troubleshooting.

Maintainability also includes assessing whether new features can be integrated into the system without causing major disruptions to existing functionalities. This flexibility allows the system to adapt to future operational needs and technological developments.

A maintainable system reduces long-term operational costs and simplifies future system improvements.

Performance

Performance measures the overall speed, responsiveness, and stability of the system during actual usage conditions. This criterion evaluates how well the system performs when handling multiple users, simultaneous reservations, and large volumes of data.

The researchers will assess the response time of the application, loading speed of pages, processing time of transactions, and overall system responsiveness. Performance testing will also determine whether the system can maintain stable operation during peak reservation periods.

The system should provide fast and smooth interactions for users without excessive delays or interruptions. Good system performance improves customer experience and operational efficiency.

Performance evaluation is important because slow or unstable systems can frustrate users and negatively affect reservation operations.

Importance of Evaluation Criteria

The use of these evaluation criteria ensures that the proposed Online Reservation and Management System for Messiah Inland Resort meets acceptable standards of quality, functionality, and usability. Through systematic evaluation, the researchers can determine whether the application effectively improves reservation management, customer service, and operational efficiency.

These criteria also help identify areas that may require further improvement or enhancement before full implementation. By evaluating the system in terms of functionality, usability, security, efficiency, reliability, accuracy, maintainability, and performance, the researchers can ensure that the developed application is suitable for practical use in the resort environment.

Overall, the evaluation process contributes to the development of a stable, secure, reliable, and user-friendly reservation system that can effectively support the operations of Messiah Inland Resort and improve the overall experience of both customers and staff.

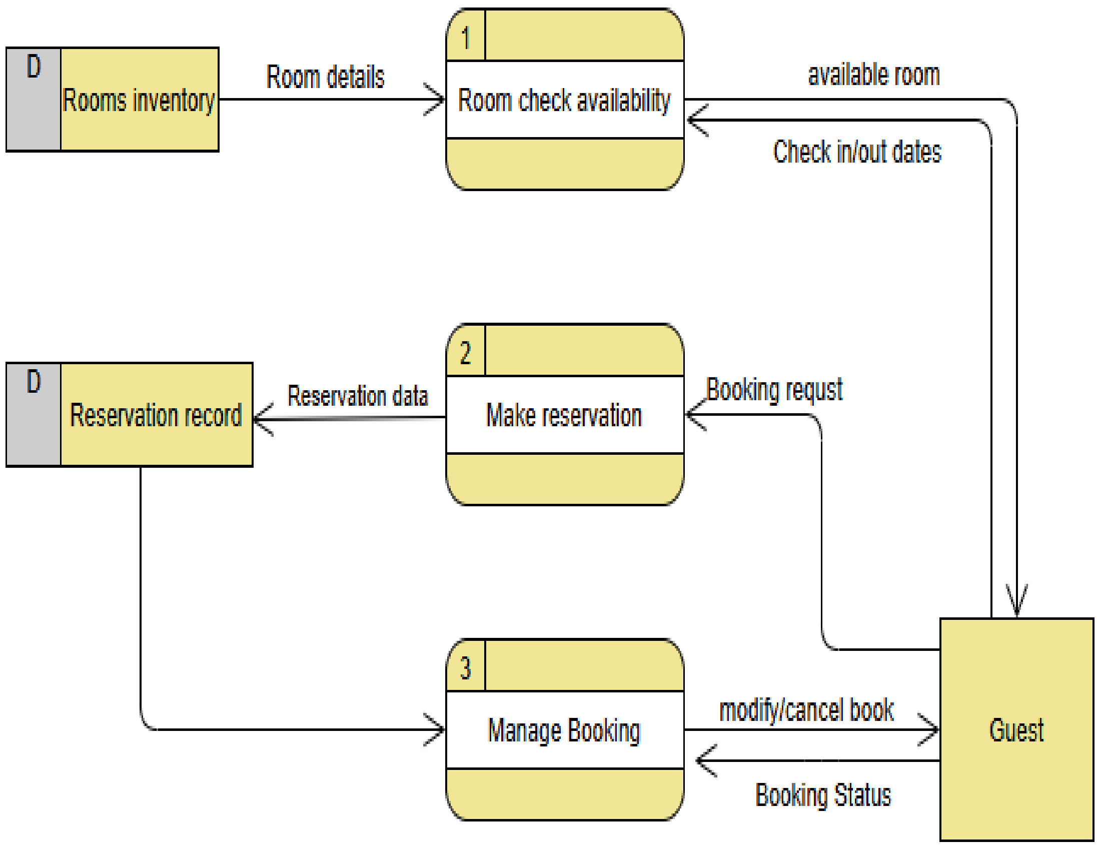
SUPPORTING DIAGRAMS (Included within Chapter 3)

Data Flow Diagram (DFD – Level 0)

User inputs data → System processes → Database stores and retrieves Information

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education



Entity Relationship Diagram (ERD)

Entities

The proposed Online Reservation and Management System for Messiah Inland Resort is composed of several important entities that represent the core components of the system. These entities are responsible for organizing, storing, processing, and managing information related to reservations, customers, payments, facilities, and resort operations.

Each entity plays a significant role in ensuring that the system functions efficiently, accurately, and securely. Together, these entities create an integrated database and management structure that supports the overall functionality of the online reservation system.

The following are the primary entities included in the proposed system:

Rooms

The Rooms entity represents the available accommodations in Messiah Inland Resort. This entity stores detailed information about each room offered by the resort and allows both customers and administrators to manage room-related data efficiently.

The Rooms entity includes information such as room type, room number, capacity, price, description, room amenities, room images, and current availability status. Additional details such as bed type, room size, air conditioning availability, Wi-Fi access, and other accommodation features may also be included.

Customers can use this entity to browse available rooms, compare accommodation options, and select rooms based on their preferences and budget. The system may display room images and descriptions to help customers make informed reservation decisions.

For administrators, the Rooms entity allows efficient management of room information, occupancy schedules, pricing updates, maintenance status, and room availability. Through centralized room management, the system helps ensure accurate reservation processing and prevents double bookings.

The Rooms entity also supports integration with the Availability Schedule and Room Maintenance entities to ensure that room status is updated in real time.

Reservation

The Reservation entity represents the booking transactions made by customers through the system. It serves as one of the most important components because it records and manages all reservation-related information.

The Reservation entity contains details such as reservation ID, booking reference number, check-in date, check-out date, number of guests, selected room or facility, reservation status, special requests, booking date, and reservation confirmation details.

Reservation status may indicate whether the booking is pending, confirmed, canceled, completed, or ongoing. This allows administrators and customers to monitor reservation progress and updates efficiently.

The Reservation entity ensures proper tracking of customer bookings and helps avoid scheduling conflicts by coordinating with the Availability Schedule entity. It also supports reservation history tracking, payment processing, and notification generation.

Accurate reservation management is essential for improving customer service and maintaining organized resort operations.

User

The User entity refers to customers who access and use the system to browse rooms, check availability, make reservations, and manage bookings.

This entity stores customer information such as full name, address, contact number, email address, username, password, and account details. Additional information such as profile images, booking preferences, and reservation history may also be included.

Users can create and manage personal accounts within the system. Through their accounts, customers can update profile information, track current and previous reservations, submit customer feedback, receive notifications, and monitor payment status.

The User entity improves customer convenience because it allows customers to manage reservations online without relying on manual booking procedures.

Proper management of user information also supports personalized services, customer communication, and reservation monitoring.

Admin

The Admin entity represents authorized resort personnel responsible for managing and maintaining the system.

Administrators have access to system management features that allow them to approve or reject reservations, update room and facility availability, manage customer accounts, process payments, handle customer concerns, and monitor resort operations.

The Admin entity stores information such as administrator name, username, password, role, contact information, and access permissions. Different administrative roles may be assigned depending on responsibilities within the resort.

Admins are also responsible for generating reports, monitoring reservation activities, updating promotional offers, and maintaining overall system performance.

This entity ensures that the system remains organized, secure, and properly managed at all times.

Facilities

The Facilities entity represents additional resort services and amenities available for reservation or customer use.

Examples of facilities include swimming pools, cottages, event spaces, recreational areas, conference halls, function rooms, playgrounds, and other resort amenities.

This entity stores details such as facility name, description, rental fee, capacity, operating hours, availability schedule, and facility status.

Customers can browse available facilities and include them in their reservations. Administrators can manage schedules, monitor facility usage, and update availability in real time.

The Facilities entity improves service management and enhances customer convenience by providing organized access to resort amenities.

Reports

The Reports entity represents generated summaries and analytical records related to resort operations and reservation activities.

This entity may include reservation reports, occupancy reports, income reports, customer activity summaries, cancellation reports, and feedback analysis.

Reports help administrators monitor business performance, identify peak booking periods, analyze customer behavior, and support decision-making processes.

The Reports entity improves operational management by providing organized and accurate information for evaluation and planning purposes.

Notifications

The Notifications entity manages automated messages, reminders, and alerts sent to customers and administrators.

Examples include reservation confirmations, payment reminders, cancellation notices, promotional announcements, booking reminders, and system alerts.

Notifications improve communication efficiency and help keep users informed about important reservation updates and transactions.

This entity also improves customer service by reducing missed bookings, delayed payments, and communication gaps.

Availability Schedule

The Availability Schedule entity tracks the real-time availability of rooms and facilities based on reservations, occupancy, cancellations, and maintenance schedules.

This entity helps prevent double booking and ensures accurate scheduling for both customers and administrators.

When reservations are made or canceled, the availability schedule automatically updates to reflect current room or facility status.

Real-time scheduling improves operational efficiency and allows customers to access updated reservation information immediately.

Customer Feedback

The Customer Feedback entity stores customer reviews, ratings, comments, and suggestions submitted after reservations or resort visits.

Customers can provide feedback regarding room quality, facilities, cleanliness, customer service, and overall satisfaction.

Administrators can analyze customer feedback to identify areas that require improvement and enhance the quality of resort services.

This entity supports continuous service improvement and customer relationship management.

Reservation History

The Reservation History entity stores previous booking records made by users, including completed, canceled, ongoing, and past reservations.

Customers can review previous transactions, while administrators can monitor booking trends and customer activities over time.

Historical reservation records are useful for generating reports, analyzing reservation patterns, and improving marketing strategies.

This entity also improves record organization and long-term reservation tracking.

Room Maintenance

The Room Maintenance entity represents maintenance schedules, repair activities, and maintenance records for rooms and resort facilities.

This entity includes information such as maintenance type, maintenance date, maintenance status, assigned personnel, and repair details.

Rooms or facilities undergoing maintenance may automatically become unavailable for reservation until repairs are completed.

The Room Maintenance entity helps ensure guest safety, cleanliness, and operational readiness of resort accommodations and facilities.

Discounts and Promotions

The Discounts and Promotions entity represents promotional offers, seasonal discounts, package deals, loyalty rewards, and coupon codes provided by the resort.

This entity stores details such as promotion name, discount percentage, validity period, eligibility requirements, and promotional codes.

The system may automatically apply eligible discounts during the reservation process.

Promotional features help attract customers, increase reservations, and improve customer retention.

Staff Management

The Staff Management entity represents resort employees and personnel responsible for handling operations and customer services.

This entity stores employee information such as names, positions, schedules, contact details, assigned responsibilities, and work assignments.

Examples of staff roles include receptionists, administrators, housekeeping personnel, maintenance workers, and security staff.

Staff Management helps improve operational coordination and supports efficient workforce management within the resort.

Importance of System Entities

The identified system entities form the foundation of the proposed Online Reservation and Management System for Messiah Inland Resort. These entities work together to support reservation processing, customer management, payment monitoring, communication, reporting, maintenance scheduling, and operational control.

By organizing information into structured entities, the system becomes more efficient, secure, reliable, and easier to manage. The integration of these entities also ensures accurate data processing, real-time updates, and improved customer service.

Overall, the system entities contribute significantly to achieving the main objective of the study, which is to develop a functional, user-friendly, and effective online reservation and management system that enhances the operational efficiency of Messiah Inland Resort.

Advantages and Benefits

- **Faster booking process** - The proposed Online Reservation and Management System provides a faster and more convenient booking process for customers and resort staff. Instead of manually recording reservations through paper forms or phone calls, customers can directly access the system online to check room availability, select their preferred room or facility, and submit their reservation in real time. This significantly reduces the time required to process bookings and minimizes delays in confirming reservations. The automated process also helps staff manage reservations more efficiently, allowing them to focus on customer service and resort operations.

- **Improved customer satisfaction** - The system enhances customer satisfaction by providing a more accessible, reliable, and user-friendly reservation experience. Customers can conveniently browse room details, view prices, check availability, and make reservations anytime and anywhere using an internet-connected device. The system also provides immediate booking confirmations and updated reservation information, helping customers feel more secure and informed about their bookings. In addition, the inclusion of customer feedback features allows users to share their experiences and suggestions, helping the resort continuously improve its services and overall guest experience.

- **Organized records** - The system helps maintain organized and centralized records of reservations, customer information, room availability, and transaction history. Instead of storing data manually in logbooks or paper files, all information is securely saved in a computerized database. This makes it easier for administrators to retrieve, update, and monitor records whenever needed. Organized records reduce the risk of lost files, duplicate entries, and inaccurate information, while also improving data accuracy and accessibility for daily resort operations.

- **Better report generation** - The Online Reservation and Management System allows administrators to generate reports quickly and accurately using stored reservation and operational data. Reports may include reservation summaries, occupancy rates, customer activity, cancellations, and income monitoring. These reports help resort management analyze business performance, identify peak booking periods, and make informed decisions regarding resort operations and customer services. Automated report generation also reduces the time and effort required to prepare manual reports and improves the reliability of information used for decision-making.

- **Reduced paperwork** - The proposed system reduces the need for manual paperwork by digitizing reservation forms, customer records, schedules, and reports. Traditional paper-based processes can consume significant time, storage space, and resources, while also increasing the risk of document loss and human error. Through the computerized system, information can be encoded, stored, updated, and retrieved electronically, making operations more efficient and environmentally friendly. Reduced paperwork also simplifies record management and improves the overall workflow of the resort staff.

- **Improved Reservation Accuracy -** The system improves reservation accuracy by automatically tracking room and facility availability in real time. This helps prevent scheduling conflicts, duplicate bookings, and incorrect reservation entries that commonly occur in manual systems. Automated validation and organized reservation monitoring ensure that booking details are accurate and properly recorded, reducing customer complaints and operational issues.

- **Enhanced Data Security -** The system provides better protection for customer and reservation data through secure user accounts and controlled administrative access. Unlike manual records that can easily be lost, damaged, or accessed by unauthorized individuals, digital records can be protected using passwords and system permissions. This improves confidentiality, reliability, and proper handling of customer information.

- **Efficient Resort Management -** The proposed system helps resort staff and administrators manage daily operations more efficiently. Tasks such as monitoring reservations, updating room availability, handling customer information, and generating reports can be completed more quickly through the centralized system. This improves productivity and allows staff to manage resort operations with less effort and fewer delays.

- **Convenient Access to Information -** Both customers and administrators can easily access important information through the system. Customers can quickly view room details, resort facilities, and booking status, while administrators can monitor occupancy, reservations, and customer activity in one centralized platform. Easy access to information improves communication, transparency, and operational efficiency.

- **Better Customer Communication -** The system improves communication between the resort and its customers by providing reservation confirmations, booking updates, and notifications. Customers can stay informed about the status of their reservations, while administrators can easily contact users regarding schedule changes, cancellations, or important announcements. Effective communication contributes to a smoother and more professional customer service experience.

Relationships:

• **Room created by Admin** – The Room entity in the proposed Online Reservation and Management System for Messiah Inland Resort is created, managed, and maintained by the Admin. Administrators play an important role in ensuring that all room information stored in the system is accurate, updated, and properly organized.

The admin is responsible for adding new room records into the system whenever new accommodations become available in the resort. During the room creation process, the admin enters important room details such as room type, room number, room capacity, pricing, room description, amenities, availability status, and room images. Additional information such as bed type, air-conditioning availability, Wi-Fi access, bathroom features, and room policies may also be included to provide customers with complete accommodation details.

Aside from creating room records, the admin is also responsible for updating and maintaining room information whenever changes occur. For example, room prices, amenities, room status, and availability schedules may be modified depending on resort operations, maintenance activities, or promotional offers. This ensures that customers always receive accurate and updated information when browsing available rooms.

The admin can also temporarily disable or mark rooms as unavailable during maintenance, repair activities, or occupancy restrictions. This feature helps prevent double bookings and ensures proper room scheduling within the system.

Through administrative management, the room entity remains organized, reliable, and synchronized with the reservation system. Proper room management also improves operational efficiency because administrators can monitor occupancy, availability, and room conditions in real time.

The Room entity is therefore directly dependent on the Admin entity because all room records and modifications are controlled and authorized by administrators. This relationship ensures accountability, accurate data management, and effective monitoring of resort accommodations within the system.

• **Reservation created by User** – The Reservation entity in the proposed Online Reservation and Management System for Messiah Inland Resort is created by the User or customer whenever they book a room, facility, or resort service through the system. Each reservation represents a formal booking transaction initiated by a registered user account.

When creating a reservation, the user selects their preferred room, facility, or accommodation based on availability, pricing, capacity, and other provided details. The customer then enters the necessary reservation information such as check-in date, check-out date, number of guests, selected facilities, special requests, and payment details if applicable.

Once the reservation is submitted, the system automatically generates a reservation record linked to the specific user account. This connection allows the system to identify which customer created the reservation and ensures proper tracking of booking activities and customer transactions.

Each reservation contains important booking information including reservation ID, booking reference number, reservation status, reservation date, payment status, and customer details. Reservation status may indicate whether the booking is pending, confirmed, canceled, completed, or ongoing.

The relationship between the User and Reservation entities is important because it allows customers to manage and monitor their own bookings through their accounts. Users can view reservation details, track reservation history, receive booking notifications, update reservation information when allowed, and check payment records.

From the administrative side, linking reservations to user accounts helps resort staff monitor customer activities, verify booking information, and provide better customer service. It also improves data organization because all reservation records are associated with specific users in the database.

The Reservation entity also interacts with other entities such as Rooms, Facilities, Payments, Availability Schedule, and Notifications to ensure accurate reservation processing and real-time updates. Once a reservation is created, the system automatically updates room or facility availability to prevent double bookings and scheduling conflicts.

Overall, the Reservation entity is directly dependent on the User entity because every reservation transaction is initiated and owned by a customer account. This relationship supports efficient reservation management, organized record keeping, and improved customer convenience within the Online Reservation and Management System for Messiah Inland Resort.

• **Admin supervises User/Booking** – In the proposed Online Reservation and Management System for Messiah Inland Resort, the Admin plays a significant role in supervising and managing both User activities and Reservation transactions. The admin is responsible for monitoring the overall operation of the system to ensure that bookings, customer records, and reservation processes function properly and efficiently.

The relationship between the Admin, User, and Reservation entities is important because it allows the resort management to maintain organized operations, monitor customer activities, and address reservation-related concerns effectively. Through administrative supervision, the system can provide reliable and accurate reservation services while minimizing operational problems and booking conflicts.

The admin monitors all user accounts registered within the system. This includes verifying customer information, managing account records, and ensuring that user activities comply with resort policies and system guidelines. Administrators may also assist users in updating account details, recovering accounts, or resolving technical issues related to login and reservation access.

In terms of reservation management, the admin supervises all booking transactions made by users. The administrator can view reservation details such as booking reference numbers, reservation dates, selected rooms or facilities, payment status, number of guests, and reservation status. This allows the admin to monitor reservation activities in real time and ensure that all bookings are processed accurately.

The admin is also responsible for approving, confirming, modifying, or canceling reservations when necessary. For example, if there are scheduling conflicts, unavailable rooms, payment issues, or special customer requests, the admin can take appropriate actions to resolve the problem and maintain smooth resort operations.

Administrative supervision helps prevent operational errors such as double bookings, incomplete reservation records, and incorrect room assignments. The admin can also monitor room availability and coordinate reservations with maintenance schedules and facility usage to ensure proper accommodation management.

Additionally, the admin handles customer concerns and booking-related inquiries. If users experience problems with reservations, payments, cancellations, or system access, administrators can provide assistance and support. This improves customer service and enhances user satisfaction.

The Admin entity also has access to reservation reports, booking histories, payment records, and customer activity logs. These monitoring features allow administrators to analyze reservation trends, identify operational issues, and evaluate resort performance more effectively.

Through the Audit Logs entity, the admin can monitor system activities such as reservation updates, cancellations, payment processing, and login attempts. This improves system security, accountability, and transparency by tracking user and administrative actions within the application.

Furthermore, administrative supervision supports the enforcement of resort policies and reservation rules. The admin can manage reservation limits, monitor occupancy schedules, update pricing information, and apply promotional offers or discounts when necessary.

The relationship between Admin, User, and Reservation entities ensures that the Online Reservation and Management System operates in an organized, secure, and efficient manner. By supervising customer activities and reservation transactions, the admin helps maintain accurate records, improve operational efficiency, and provide reliable services to customers.

Overall, the Admin entity serves as the central management authority of the system, ensuring that all user interactions and booking processes are properly monitored and controlled for the successful operation of Messiah Inland Resort.

• **User cancels Reservation** – In the proposed Online Reservation and Management System for Messiah Inland Resort, users are given the ability to cancel their reservations when necessary, subject to the resort's cancellation policies, rules, and system conditions. The cancellation feature provides flexibility and convenience for customers while helping the resort maintain organized reservation management.

The relationship between the User and Reservation entities allows customers to manage their own bookings through their registered accounts. Users can access their reservation details, review booking information, and request cancellation directly within the system without requiring manual processing from resort staff.

When a user decides to cancel a reservation, the system verifies whether the booking qualifies for cancellation based on the resort's established policies. These policies may include cancellation deadlines, refund conditions, cancellation fees, and restrictions for promotional or discounted bookings.

For example, the resort may allow free cancellation within a specified number of days before the scheduled check-in date. Reservations canceled beyond the allowed cancellation period may be subject to penalties, partial refunds, or non-refundable conditions depending on resort policies.

Once the cancellation request is approved, the system automatically updates the reservation status from confirmed or pending to canceled. The canceled reservation is then recorded in the Reservation History entity for future reference and monitoring purposes.

The cancellation process also updates the Availability Schedule entity in real time. Rooms or facilities associated with the canceled reservation become available again for booking by other customers. This feature helps maximize room occupancy and prevents scheduling conflicts.

If payment has already been processed, the system may also coordinate with the Payments entity to determine whether a refund, balance adjustment, or cancellation fee applies. Administrators can review payment records and process refunds according to resort financial policies.

Notifications are also generated during the cancellation process. Customers may receive automated cancellation confirmations through email, SMS, or in-system notifications. Administrators may also receive alerts regarding canceled bookings to ensure proper monitoring of reservation activities.

The admin retains the authority to review, approve, or manage cancellation requests when necessary, especially in cases involving refund disputes, policy exceptions, or reservation modifications. This administrative supervision ensures that cancellation procedures remain fair, organized, and aligned with resort policies.

The ability for users to cancel reservations improves customer convenience and flexibility because customers can manage unexpected schedule changes or personal circumstances more easily. It also enhances customer satisfaction by providing transparent and accessible reservation management options.

At the same time, implementing cancellation rules within the system helps protect the operational and financial interests of the resort. Controlled cancellation procedures reduce sudden booking losses, improve room management, and maintain organized scheduling.

Overall, the User Cancels Reservation relationship demonstrates the interaction between the User and Reservation entities within the Online Reservation and Management System. This feature supports efficient booking management, accurate availability updates, customer convenience, and proper enforcement of resort reservation policies.

• **User makes Reservation for Room** – In the proposed Online Reservation and Management System for Messiah Inland Resort, a User can create a reservation for one or more available rooms based on their accommodation needs, preferences, and the current room availability within the resort. This relationship represents one of the primary functions of the system because it allows customers to conveniently book rooms through an online platform.

The reservation process begins when a user accesses the system using a registered account. Through the user interface, customers can browse available rooms, view room descriptions, compare prices, examine room amenities, and check availability schedules in real time. The system provides detailed information about each room to help users select accommodations that best fit their requirements.

Once the user selects a room, the system verifies whether the room is available for the chosen check-in and check-out dates. The Availability Schedule entity automatically checks existing reservations, occupancy records, and maintenance schedules to ensure that the selected room can still be reserved.

Users may reserve one or multiple rooms depending on the number of guests, accommodation preferences, or booking requirements. For example, families, large groups, or event participants may reserve several rooms under a single reservation transaction. The system allows multiple room selections while maintaining accurate booking records and availability updates.

During the reservation process, the user provides important booking details such as:

- Check-in date
- Check-out date
- Number of guests
- Selected room or room type
- Special requests or preferences
- Contact information
- Payment details if required

After the reservation information is submitted, the system creates a Reservation record linked to the specific User account and selected Room entity. The system then generates a booking reference number and updates the room status to reserved or unavailable for the selected dates.

The reservation information is automatically stored in the database to ensure accurate record keeping and future reference. The user may also receive automated notifications confirming the reservation details, payment status, and booking schedule.

The relationship between the User, Reservation, and Room entities ensures proper coordination of room bookings and availability management. Once a reservation is successfully completed, the Availability Schedule entity updates in real time to prevent double bookings and scheduling conflicts.

From the administrative side, resort staff and administrators can monitor room reservations, occupancy levels, customer booking activities, and reservation statuses through the management dashboard. This helps improve operational efficiency and ensures organized room allocation.

The ability for users to reserve one or multiple rooms online improves customer convenience because bookings can be completed anytime and anywhere using internet-connected devices. Customers no longer need to rely solely on manual reservation methods such as phone calls or walk-in bookings.

Additionally, this feature improves operational efficiency within the resort by automating reservation processing, reducing manual errors, and providing accurate room availability information in real time.

Overall, the User Makes Reservation for Room relationship demonstrates the interaction between the User, Reservation, and Room entities within the Online Reservation and Management System. This relationship supports efficient online booking, organized reservation management, accurate availability monitoring, and improved customer service for Messiah Inland Resort.

• **Reservation assigned to Room** – In the proposed Online Reservation and Management System for Messiah Inland Resort, each Reservation is assigned to a specific Room to ensure accurate booking management, occupancy monitoring, and scheduling control. This relationship between the Reservation and Room entities is essential because it allows the system to track which rooms are reserved, occupied, available, or unavailable at any given time.

When a user creates a reservation, the system automatically links the reservation record to the selected room or rooms chosen during the booking process. This assignment ensures that the reservation information is directly associated with the corresponding accommodation within the resort.

The Reservation entity contains important booking details such as check-in date, check-out date, number of guests, reservation status, and booking reference number. Meanwhile, the Room entity stores information about the room itself, including room type, room number, price, amenities, capacity, and availability status. By connecting these two entities, the system can manage reservations more efficiently and maintain accurate room schedules.

Once a reservation is assigned to a room, the system updates the room's availability status in real time. The selected room becomes unavailable for booking during the reserved dates to prevent double bookings and scheduling conflicts. This automatic update ensures that customers only see rooms that are currently available when making reservations.

The assignment process also helps administrators monitor room occupancy effectively. Resort staff can easily identify which rooms are occupied, reserved, vacant, under maintenance, or scheduled for future bookings. This improves operational organization and supports proper accommodation management.

In cases where multiple rooms are reserved under a single booking transaction, the system may create separate room assignments linked to one reservation record. This setup allows the system to handle group bookings, family accommodations, and event-related reservations more efficiently.

The Reservation assigned to Room relationship also supports accurate billing and payment processing. Since each reservation is connected to a specific room, the system can calculate accommodation fees based on room type, pricing, duration of stay, and additional services included in the booking.

Furthermore, this relationship improves reporting and reservation tracking. Administrators can generate reports related to room occupancy rates, booking schedules, reservation histories, and room utilization patterns. These reports help management analyze resort performance and improve operational planning.

The assignment of reservations to rooms also integrates with the Availability Schedule and Room Maintenance entities. If a room is under maintenance or temporarily unavailable, the system prevents it from being assigned to new reservations. Similarly, canceled reservations automatically release room assignments and update room availability for future bookings.

For customers, this feature improves reservation accuracy and transparency because users can clearly identify the rooms they reserved along with the corresponding booking schedules and accommodation details.

Overall, the Reservation assigned to Room relationship plays an important role in ensuring organized reservation management, accurate occupancy monitoring, real-time availability updates, and efficient resort operations within the Online Reservation and Management System for Messiah Inland Resort.

• **Admin manages Reservation status** – In the proposed Online Reservation and Management System for Messiah Inland Resort, the Admin is responsible for managing the status of all reservations within the system. This relationship between the Admin and Reservation entities is important because it ensures that booking transactions are properly monitored, updated, and controlled according to resort policies, room availability, and operational requirements.

The admin has the authority to approve, confirm, modify, update, or cancel reservations depending on the current status of accommodations and the specific needs of customers. Through administrative management, the resort can maintain organized booking operations and provide reliable reservation services.

When a customer creates a reservation, the system may initially mark the booking as pending until it is reviewed by the admin. The administrator checks important reservation details such as room availability, payment status, booking dates, number of guests, and customer requests before approving the reservation.

Once verified, the admin can update the reservation status from pending to confirmed. Confirmation indicates that the room or facility has been successfully reserved for the customer and that the booking meets the resort’s operational requirements and policies.

In situations where rooms or facilities are unavailable, incomplete payment information is provided, or reservation conflicts occur, the admin may reject or cancel the reservation. This ensures that invalid or problematic bookings do not affect resort operations or create scheduling issues.

The admin can also modify reservation information when necessary. Changes may include updating check-in or check-out dates, changing room assignments, adjusting guest details, or adding special accommodations requested by customers. These updates help provide flexibility and improve customer service.

Reservation management is closely connected with the Availability Schedule entity. Whenever the admin approves, updates, or cancels a reservation, the system automatically updates room or facility availability in real time. This prevents double booking and ensures accurate scheduling throughout the resort.

The admin is also responsible for monitoring canceled reservations and applying resort cancellation policies when needed. If a reservation is canceled, the administrator may coordinate with the Payments entity to process refunds, penalties, or payment adjustments depending on the resort's rules and conditions.

Additionally, the admin can monitor reservation statuses such as:

- Pending
- Confirmed
- Ongoing
- Completed
- Canceled
- Rejected

These status categories help organize reservation records and improve operational monitoring. Administrators can easily track current bookings, upcoming reservations, and completed transactions within the system.

The management of reservation statuses also improves communication between the resort and customers. Once changes are made to a reservation, the system may automatically send notifications to users informing them about booking confirmations, cancellations, payment updates, or schedule modifications.

Administrative supervision ensures that all reservations comply with resort policies, occupancy limits, maintenance schedules, and operational procedures. It also allows the resort to respond quickly to booking concerns, customer requests, and scheduling conflicts.

Furthermore, managing reservation statuses helps improve reporting and monitoring functions. Administrators can generate reservation summaries, occupancy reports, cancellation records, and booking statistics that support decision-making and business analysis.

The relationship between the Admin and Reservation entities is therefore essential for maintaining efficient reservation processing, accurate room management, and organized resort operations. By supervising and controlling reservation statuses, the admin helps ensure reliable customer service, operational stability, and proper implementation of resort policies.

Overall, the Admin manages Reservation status relationship demonstrates the administrator’s role in controlling and maintaining reservation activities within the Online Reservation and Management System for Messiah Inland Resort.

• **Room has Availability status** – In the proposed Online Reservation and Management System for Messiah Inland Resort, each Room has an Availability Status that indicates whether the room is currently available, reserved, occupied, under maintenance, or temporarily unavailable. This relationship is important because it allows the system to monitor room conditions and booking schedules accurately in real time.

The availability status of a room is continuously updated based on reservation activities, occupancy records, cancellations, maintenance schedules, and administrative actions. Through this feature, the system ensures that customers only view rooms that are actually available for reservation, helping prevent booking conflicts and operational errors.

The Room entity stores detailed accommodation information such as room type, capacity, pricing, amenities, and room description. Along with these details, the system also maintains the room’s current status to support accurate reservation management and occupancy monitoring.

The most common room availability statuses include:

- Available – Indicates that the room is open for reservation and currently not assigned to any guest or booking schedule.
- Reserved – Indicates that the room has already been booked by a customer for specific check-in and check-out dates but is not yet occupied.
- Occupied – Indicates that the room is currently being used by guests during their stay at the resort.
- Under Maintenance – Indicates that the room is temporarily unavailable due to cleaning, repair, inspection, or maintenance activities.
- Unavailable – Indicates that the room cannot currently be reserved because of operational restrictions or administrative actions.

When a customer creates a reservation, the system automatically checks the Availability Schedule entity to determine whether the selected room is available during the requested dates. Once the reservation is confirmed, the room status changes from available to reserved.

On the customer's check-in date, the room status may automatically update from reserved to occupied to reflect actual room usage. After the customer checks out and housekeeping or inspections are completed, the room status may return to available.

If a reservation is canceled, the system immediately updates the room's availability status and makes the room available again for future bookings. This real-time synchronization improves operational efficiency and maximizes room utilization within the resort.

The admin also has the authority to manually update room availability statuses when necessary. For example, rooms undergoing maintenance or repairs can be temporarily marked as unavailable or under maintenance to prevent reservations from being accepted during repair periods.

The Room has Availability Status relationship is closely connected to several system entities including Reservation, Availability Schedule, Room Maintenance, and Admin. These entities work together to ensure that room occupancy and booking schedules remain accurate and properly coordinated.

This relationship also supports efficient reporting and monitoring functions. Administrators can easily track room occupancy rates, identify vacant accommodations, monitor maintenance schedules, and analyze booking trends through real-time room status information.

For customers, accurate availability status improves booking convenience and transparency because users can immediately identify which rooms are open for reservation. Real-time updates also reduce frustration caused by outdated or incorrect room availability information.

From an operational perspective, continuous room status monitoring helps improve reservation accuracy, prevent double bookings, and maintain organized accommodation management. Resort staff can efficiently coordinate check-ins, check-outs, housekeeping activities, and maintenance schedules based on room availability information.

Overall, the Room has Availability Status relationship plays a vital role in ensuring accurate reservation processing, organized room management, and smooth resort operations within the Online Reservation and Management System for Messiah Inland Resort.

• **User receives Reservation confirmation** – In the proposed Online Reservation and Management System for Messiah Inland Resort, the User receives a Reservation Confirmation after successfully completing a booking transaction. This relationship is important because it provides customers with official verification that their reservation has been processed and recorded by the system.

The reservation confirmation serves as proof of booking and contains important details regarding the customer's reservation. It helps ensure that both the customer and the resort have accurate and organized records of the transaction. Confirmation messages also improve customer confidence and satisfaction because users immediately know that their reservation request has been successfully accepted.

When a user submits a reservation request through the system, the booking information is processed and verified based on room availability, reservation schedules, and system requirements. Once the reservation is approved or successfully recorded, the system automatically generates a reservation confirmation linked to the user's account and booking record.

The reservation confirmation typically includes important information such as:

- Reservation reference number
- Customer name
- Selected room or facility
- Check-in and check-out dates
- Number of guests
- Reservation status
- Payment status
- Total booking amount
- Special requests or notes
- Resort contact information

The system may send the confirmation through different communication methods such as email notifications, SMS messages, or in-system notifications. This ensures that customers can easily access and review their booking details anytime.

The reservation confirmation feature is closely connected to the Notifications entity because automated alerts and confirmation messages are generated whenever a reservation is successfully completed. Notifications help improve communication between the resort and customers by providing timely updates regarding bookings and reservation changes.

The confirmation process also helps minimize misunderstandings and booking disputes because customers can verify the accuracy of their reservation details immediately after booking. If errors or concerns are identified, customers can contact the resort or request modifications before their scheduled check-in date.

From the administrative side, reservation confirmations help maintain organized transaction records and support efficient reservation tracking. Administrators can review confirmation logs, monitor reservation activities, and verify customer bookings more effectively.

Additionally, reservation confirmations improve operational efficiency because they reduce the need for manual follow-ups and repeated booking inquiries. Customers can independently access their reservation information using the confirmation details provided by the system.

The reservation confirmation may also include reminders regarding resort policies, cancellation rules, payment instructions, and check-in procedures. Providing these details helps customers prepare for their stay and ensures that they understand the resort's booking guidelines.

In some cases, the system may generate updated confirmations if reservation details are modified by the customer or admin. For example, changes in room assignments, reservation dates, or payment status may trigger new confirmation notifications to keep customers informed about updated booking information.

The relationship between the User and Reservation Confirmation entities supports transparency, accountability, and customer convenience within the system. It ensures that customers receive accurate booking verification while helping the resort maintain organized reservation records and communication processes.

Overall, the User receives Reservation confirmation relationship plays an important role in improving customer experience, reservation accuracy, and communication efficiency within the Online Reservation and Management System for Messiah Inland Resort.

• **Admin generates Reports from Reservations** – In the proposed Online Reservation and Management System for Messiah Inland Resort, the Admin has the ability to generate reports based on reservation data and other operational records stored within the system. This relationship between the Admin and Reports entities is important because it helps management monitor resort activities, analyze business performance, and support effective decision-making processes.

Reports generated from reservation data provide organized and summarized information regarding customer bookings, room occupancy, facility usage, payment transactions, cancellations, and overall system activity. These reports allow administrators to evaluate the efficiency of resort operations and identify important trends related to customer behavior and reservation patterns.

The admin can access the reporting feature through the administrative dashboard of the system. Using this feature, administrators can generate different types of reports depending on the information needed for monitoring and management purposes.

Reservation reports contain important booking information such as reservation dates, customer names, room assignments, booking statuses, payment records, and check-in or check-out schedules. These reports help administrators track current and upcoming reservations accurately.

Occupancy reports allow the admin to monitor how frequently rooms and facilities are used within a certain period. This information helps management identify peak booking seasons, high-demand accommodations, and underutilized facilities.

Peak season analysis is especially important because it helps the resort prepare for periods of increased customer demand. By reviewing reservation trends and occupancy rates, administrators can improve staffing schedules, room management, promotional planning, and resource allocation during busy seasons.

The admin can also generate cancellation reports to monitor canceled reservations and identify common cancellation patterns. These reports help management evaluate the effectiveness of cancellation policies and identify operational issues that may affect customer bookings.

Revenue and payment reports provide financial summaries related to reservation transactions. These reports help administrators monitor income, unpaid balances, payment statuses, and overall resort earnings. Financial reports also support accounting and budgeting activities.

Customer activity reports allow the admin to analyze user behavior such as frequent bookings, preferred room types, reservation history, and customer feedback. This information can help the resort improve customer service and develop better marketing strategies.

The report generation process is closely connected with several system entities including Reservation, User, Payments, Rooms, Facilities, and Reservation History. The system collects and organizes data from these entities to produce accurate and updated reports.

Reports may be generated in different formats such as tables, summaries, printable documents, or downloadable files depending on the system design. Some reports may also include charts or graphical representations to help administrators analyze trends more effectively.

The ability to generate reports improves operational efficiency because administrators no longer need to manually organize reservation records or calculate occupancy information. Automated reporting reduces human error, saves time, and provides faster access to important business data.

In addition, report generation supports strategic planning and decision-making. Administrators can use reservation statistics and operational reports to evaluate resort performance, identify business opportunities, improve customer services, and plan future system improvements.

The Reports entity also helps improve accountability and transparency because all reservation activities and transactions are properly documented and accessible for review. Administrators can monitor system usage and operational performance more effectively using accurate and organized reports.

Overall, the Admin generates Reports from Reservations relationship demonstrates the important role of administrative reporting within the Online Reservation and Management System for Messiah Inland Resort. Through automated report generation, the system supports efficient monitoring, operational analysis, informed decision-making, and improved management of resort services and reservation activities.

• **Customer Feedback** – In the proposed Online Reservation and Management System for Messiah Inland Resort, the Customer Feedback entity allows users to provide reviews, ratings, comments, and suggestions after completing a reservation or staying at the resort. This relationship is important because it helps the resort evaluate customer satisfaction, improve service quality, and enhance the overall guest experience.

Customer feedback serves as a valuable source of information for both administrators and resort management. Through customer opinions and experiences, the resort can identify strengths, weaknesses, and areas that require improvement in terms of accommodations, facilities, services, and operational processes.

After a reservation is completed or a guest checks out from the resort, the system may allow the user to access a feedback form through their account. Customers can then submit their evaluation regarding different aspects of their stay and overall experience.

The Customer Feedback entity stores all submitted reviews and comments in the system database for future reference and analysis. Each feedback record may include customer information, reservation details, rating scores, feedback date, and written remarks.

The relationship between the User, Reservation, and Customer Feedback entities ensures that only customers with completed or valid reservations can submit reviews. This helps maintain the authenticity and reliability of customer evaluations within the system.

Customer feedback provides important benefits to the resort management. Administrators can review comments and ratings to identify common customer concerns, monitor service performance, and evaluate employee efficiency. Negative feedback may help management identify operational problems such as delayed services, maintenance issues, booking difficulties, or facility concerns.

Positive feedback, on the other hand, helps the resort recognize areas where services are performing well and encourages staff to maintain high-quality customer service. Feedback may also be used for marketing purposes by showcasing positive customer experiences and satisfaction levels.

The admin may monitor customer reviews through the administrative dashboard and generate reports related to customer satisfaction trends, service evaluations, and frequently mentioned concerns. These reports help management make informed decisions regarding operational improvements and customer service strategies.

The Customer Feedback entity also supports communication between the resort and its customers. Administrators may respond to customer concerns, acknowledge suggestions, or address complaints appropriately. This interaction helps improve customer relationships and demonstrates the resort’s commitment to service improvement.

Additionally, feedback information may help guide future business decisions such as facility upgrades, staff training, promotional planning, and service enhancements. Understanding customer preferences and expectations allows the resort to provide better experiences for future guests.

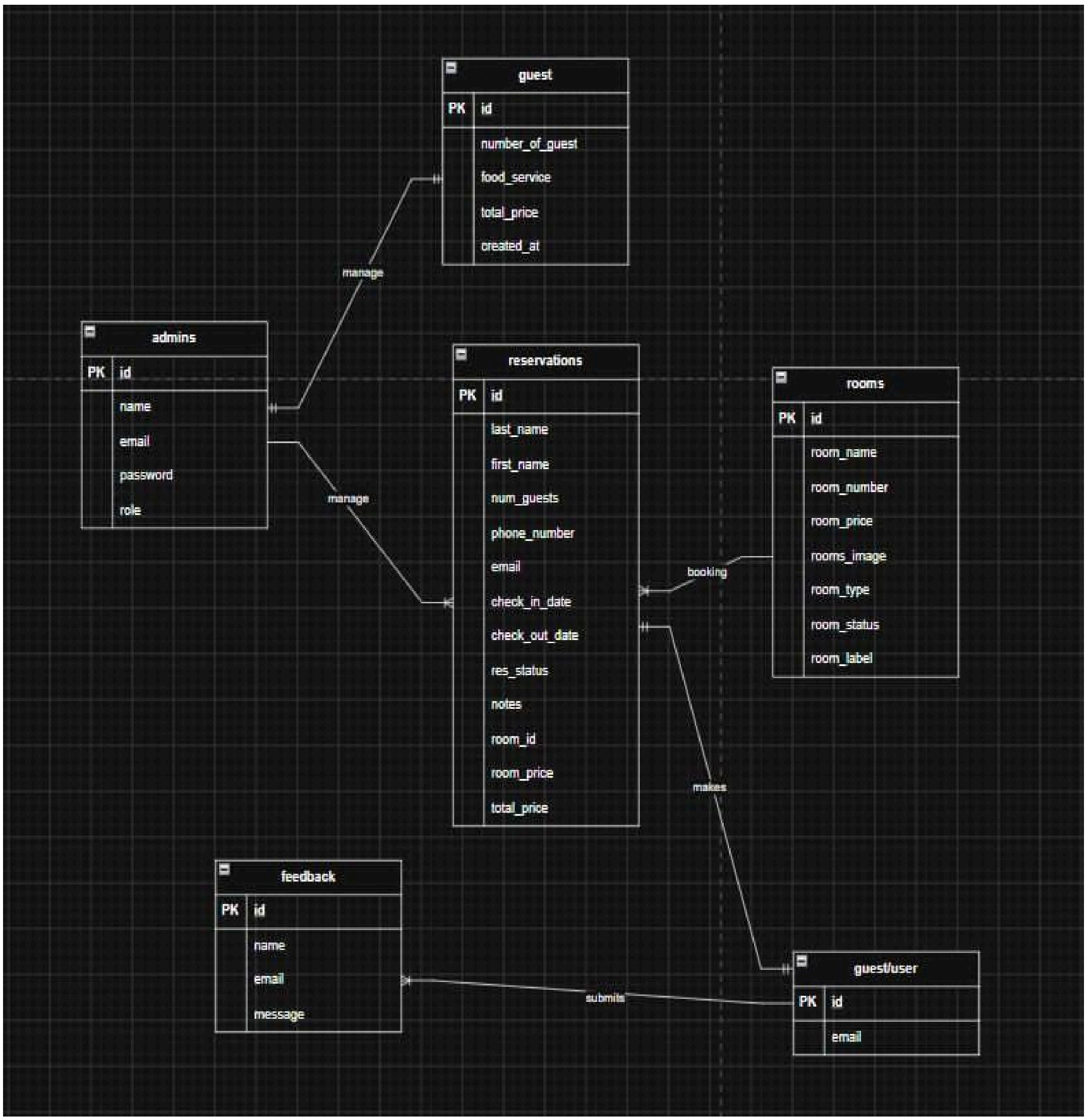
The integration of customer feedback within the online reservation system improves transparency and accountability because management can continuously evaluate resort performance based on actual guest experiences.

From the customer’s perspective, the ability to provide feedback gives users an opportunity to express their opinions and contribute to the improvement of resort services. Customers may feel more valued when their suggestions and concerns are acknowledged by management.

Overall, the Customer Feedback relationship plays an important role in improving service quality, customer satisfaction, operational evaluation, and continuous development within the Online Reservation and Management System for Messiah Inland Resort. By collecting and analyzing customer feedback, the resort can enhance guest experiences and maintain high standards of hospitality and customer service.

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education



3.10 USE CASE DIAGRAM

Actors:

- **Guest (Customer)** – The main user who interacts with the system to view available rooms, check availability, and make reservations. Guests can also receive booking confirmations and manage their own reservations.

- **Administrator** – The authorized user responsible for managing the entire system, including room management, reservation approval, user accounts, and system monitoring.

- **Receptionist / Staff** – Assists in handling reservations, updating booking records, checking room availability, and assisting guests with inquiries.

- **System** – Represents the automated processes of the system such as sending notifications, updating availability in real time, and generating reports.

- **Manager / Owner** – Oversees overall resort operations, views reports, and monitors reservation trends and system performance for decision-making.

- **Registered User** – A guest who has created an account in the system and can access additional features such as booking history and reservation management.

These actors interact with the Online Reservation and Management System for Messiah Inland Resort to perform various functions that improve reservation handling, data management, and overall resort operations.

Uses Case:

USE CASES

- **View Resort Information** – In the proposed Online Reservation and Management System for Messiah Inland Resort, the View Resort Information feature allows guests and users to access important details about the resort through the online platform. This feature serves as one of the primary functions of the system because it provides customers with the necessary information needed before making reservations or inquiries.

The ability to view resort information improves customer convenience and enhances the overall booking experience by allowing users to explore accommodations, facilities, services, and pricing without needing to visit the resort personally. Through this feature, customers can make informed decisions regarding room selection, facility reservations, and travel planning.

The Room entity is closely connected to this feature because customers can view room descriptions, room images, accommodation details, occupancy limits, and current availability status. Real-time room availability helps users identify which rooms can be reserved during their preferred dates.

The Facilities entity also supports this feature by displaying information about resort amenities such as swimming pools, function halls, recreational spaces, cottages, event venues, and other services available for guests. Customers can review facility details including rental fees, capacity limits, operating schedules, and usage guidelines.

Pricing information is also presented to customers to ensure transparency in reservation transactions. Users can compare room rates, facility charges, and promotional packages before finalizing reservations. This helps customers select accommodations that match their preferences and budget.

The View Resort Information feature may also include image galleries and descriptions to help customers visualize the resort environment, accommodations, and amenities. Providing visual content improves customer engagement and increases the attractiveness of the resort’s online platform.

In addition, the system may display resort policies such as cancellation rules, payment procedures, check-in and check-out schedules, safety guidelines, and reservation conditions. This helps ensure that customers understand the resort’s operational policies before making bookings.

The feature also supports promotional and marketing activities by allowing administrators to display seasonal offers, discounts, package deals, and special events. Promotional information helps attract more customers and encourages reservations during both peak and off-peak seasons.

From the administrative perspective, the admin manages and updates all resort information displayed within the system. Administrators can add new room details, modify prices, update facility information, upload images, and announce promotional offers as needed. This ensures that customers always receive accurate and updated information.

The View Resort Information feature improves communication between the resort and potential customers by providing centralized access to important resort details. Customers no longer need to rely solely on phone inquiries or manual visits to gather information about accommodations and services.

Additionally, this feature improves operational efficiency because it reduces repetitive customer inquiries regarding room availability, pricing, and resort services. Customers can independently access the information they need directly through the online system.

The relationship between the User, Room, Facilities, and Admin entities supports the proper operation of this feature. Users access and review resort information, while administrators manage and maintain the displayed content within the system database.

Overall, the View Resort Information feature plays an important role in improving customer convenience, promoting resort services, supporting reservation decisions, and enhancing the overall functionality of the Online Reservation and Management System for Messiah Inland Resort. By providing accessible and updated resort information, the system helps create a more efficient, informative, and user-friendly reservation experience for guests and customers.

- **Inquire/Book Room** – In the proposed Online Reservation and Management System for Messiah Inland Resort, the Inquire/Book Room feature allows guests and users to check room availability and make reservations online by selecting their preferred dates, room types, and accommodation options. This feature is one of the primary functions of the system because it provides customers with a convenient and efficient method of reserving rooms without relying on manual booking procedures.

The Inquire/Book Room feature simplifies the reservation process by allowing customers to access booking services anytime and anywhere through an internet-connected device. Users no longer need to visit the resort personally or contact staff manually to inquire about room availability and reservation details.

Customers can compare different room options based on their accommodation preferences, number of guests, budget, and desired amenities. This helps users make informed reservation decisions more easily.

The inquiry process allows users to check room availability by entering preferred check-in and check-out dates. The system automatically verifies room schedules through the Availability Schedule entity to determine whether rooms are available during the selected dates.

If rooms are available, the system displays matching accommodation options for the user. If no rooms are available, the system may suggest alternative dates, room types, or available facilities depending on the system design.

After submitting the reservation request, the system creates a Reservation record linked to the selected room and the user's account. A unique booking reference number is automatically generated for tracking and verification purposes.

The system then updates the room's availability status in real time to prevent double bookings and scheduling conflicts. Reserved rooms become temporarily unavailable for other customers during the selected booking period.

The Inquire/Book Room feature is closely connected to several system entities including User, Room, Reservation, Payments, Availability Schedule, and Notifications. These entities work together to ensure accurate reservation processing and smooth booking operations.

The system may also generate automated reservation confirmations and notifications once the booking is successfully processed. Customers may receive confirmation messages through email, SMS, or in-system notifications containing important booking details and payment instructions.

From the administrative perspective, resort staff and administrators can monitor all room inquiries and reservations through the administrative dashboard. Admins can approve, modify, confirm, or cancel bookings depending on room availability, payment status, and resort policies.

The online inquiry and booking feature significantly improves operational efficiency because it automates reservation processing, reduces manual paperwork, minimizes booking errors, and provides real-time availability updates.

This feature also enhances customer convenience and satisfaction because guests can complete reservations quickly and independently without waiting for manual confirmation through phone calls or in-person visits.

Additionally, the Inquire/Book Room feature helps improve resort occupancy management by allowing administrators to monitor booking trends, identify peak reservation periods, and optimize room utilization.

Overall, the Inquire/Book Room feature plays an essential role in the Online Reservation and Management System for Messiah Inland Resort. It supports efficient reservation management, real-time availability monitoring, improved customer service, and organized booking operations while providing guests with a faster and more convenient reservation experience.

- **Approve Bookings (Admin)** – In the proposed Online Reservation and Management System for Messiah Inland Resort, the Approve Bookings feature allows the administrator to review, approve, reject, or manage reservation requests submitted by guests through the online booking system. This feature is important because it ensures that all reservations are properly verified and processed according to room availability, payment status, resort policies, and operational requirements.

The approval process begins when a guest submits a reservation request through the system. Once the booking information is entered, the reservation is temporarily recorded with a pending status while awaiting administrative review and confirmation.

The administrator carefully reviews each reservation request to verify whether the selected room or facility is available during the requested schedule. The system also checks the Availability Schedule entity to identify existing bookings, maintenance schedules, or occupancy conflicts.

If the reservation satisfies all requirements and the selected room is available, the admin may approve and confirm the booking. Once approved, the system automatically updates the reservation status from pending to confirmed. At the same time, the room's availability status changes to reserved to prevent double bookings and scheduling conflicts.

When a booking is declined, the system may notify the customer regarding the reason for the rejection and may suggest alternative accommodations or schedules if available.

The administrator may also modify reservation details when necessary. Changes may include updating reservation dates, changing room assignments, adjusting guest information, or responding to special customer requests. This flexibility improves customer service and allows the resort to accommodate booking adjustments efficiently.

The Approve Bookings feature is closely connected to several system entities including Reservation, Room, User, Payments, Availability Schedule, and Notifications. These entities work together to ensure accurate reservation processing and proper booking management.

Once a reservation is approved or declined, the system automatically sends notifications to the customer. Confirmation messages may include booking details, payment instructions, check-in schedules, and reservation policies. Declined reservations may also generate notification messages explaining the status update.

Administrative approval helps maintain organized reservation records and improves operational efficiency because all bookings are monitored and verified before final confirmation. This reduces errors such as double bookings, incorrect room assignments, and incomplete reservations.

The approval process also supports security and accountability within the system because only authorized administrators can approve or reject reservation transactions. Administrative actions may be recorded in the Audit Logs entity to track booking decisions and system activities.

From a business perspective, the Approve Bookings feature helps resort management maintain control over room occupancy, reservation schedules, and customer accommodations. Administrators can prioritize reservations, monitor occupancy trends, and coordinate bookings with maintenance schedules and resort operations.

Additionally, this feature enhances customer trust and satisfaction because guests receive verified and organized booking confirmations directly from the resort management.

Overall, the Approve Bookings (Admin) feature plays an essential role in ensuring accurate reservation processing, efficient room management, operational control, and reliable customer service within the Online Reservation and Management System for Messiah Inland Resort.

- **Monitor Analytics (Admin) -** In the proposed Online Reservation and Management System for Messiah Inland Resort, the Monitor Analytics feature provides administrators with access to important reports, summaries, and statistical data related to reservations, customer activities, room occupancy, and overall resort operations. This feature helps the admin analyze business performance, monitor operational efficiency, and support strategic decision-making processes.

The analytics and reporting functions allow administrators to gather meaningful insights from the data collected within the system. Through organized reports and summarized information, the admin can evaluate reservation trends, identify peak booking periods, monitor customer behavior, and assess the effectiveness of resort services and operations.

The Monitor Analytics feature is important because it transforms raw reservation and transaction data into useful information that can support management decisions and operational improvements. By analyzing reservation statistics and customer activities, administrators can better understand the resort's performance and plan future business strategies.

Reservation statistics provide administrators with information regarding the total number of bookings, completed reservations, canceled bookings, pending reservations, and ongoing stays within a specific period. These reports help management monitor reservation activity and evaluate booking performance.

Booking trend analysis allows the admin to identify periods when reservations increase or decrease. This information is useful in recognizing seasonal demand patterns, holidays, special events, and high-traffic booking periods.

Peak season reports are especially valuable because they help administrators prepare for periods of increased customer demand. By identifying peak booking months or seasons, the resort can improve room allocation, staff scheduling, inventory preparation, and operational planning.

Room occupancy reports allow the admin to monitor how frequently rooms are being reserved and occupied. These reports help determine which accommodations are most popular and which rooms may require promotional strategies to improve occupancy rates.

Revenue and payment analytics provide financial summaries related to reservation transactions, payments received, outstanding balances, refunds, and overall resort income. Financial reports help management evaluate profitability and support budgeting decisions.

The Monitor Analytics feature also allows administrators to analyze customer behavior and preferences. Customer activity reports may include information regarding frequent guests, preferred room types, booking habits, reservation frequency, and customer feedback trends.

Customer feedback analytics help management evaluate customer satisfaction levels and identify areas requiring service improvement. By analyzing reviews, ratings, and suggestions, administrators can improve resort facilities, customer service, and operational efficiency.

The analytics system is closely connected to several entities including Reservation, User, Room, Payments, Facilities, Customer Feedback, and Reservation History. Data collected from these entities is processed and organized into meaningful reports and visual summaries.

The system may also display analytics using tables, charts, graphs, or dashboards to help administrators understand reservation patterns and operational trends more easily. Visual representations improve data interpretation and support faster decision-making.

The Monitor Analytics feature significantly improves operational efficiency because administrators no longer need to manually compute reservation statistics or organize booking records. Automated analytics reduce human error, save time, and provide quick access to updated business information.

In addition, analytics monitoring supports long-term planning and business development. Resort management can use analytical reports to create marketing strategies, improve customer services, develop promotional offers, optimize staffing schedules, and increase operational productivity.

The feature also enhances accountability and transparency because all reservation transactions and operational records are properly documented and available for administrative review.

From a managerial perspective, monitoring analytics helps administrators identify operational problems, evaluate system performance, and make informed business decisions based on accurate and real-time information.

Overall, the Monitor Analytics (Admin) feature plays an essential role in improving operational monitoring, business analysis, reservation management, and strategic planning within the Online Reservation and Management System for Messiah Inland Resort. Through analytics and reporting tools, administrators can effectively monitor resort performance, improve services, and support efficient resort operations.

- **Manage Room Availability** – In the proposed Online Reservation and Management System for Messiah Inland Resort, the Manage Room Availability feature allows the administrator to add, update, remove, and monitor room information while maintaining accurate real-time availability status for all resort accommodations. This feature is essential because it ensures that room records remain updated, organized, and synchronized with reservation activities and operational requirements.

The management of room availability is one of the most important responsibilities of the administrator because it directly affects reservation accuracy, customer satisfaction, and the overall efficiency of resort operations. Through this feature, the admin can control how room information is displayed and managed within the system.

The admin accesses the room management functions through the administrative dashboard, where all room records and availability schedules are stored and monitored. The system allows the administrator to perform several room management tasks including:

- Adding new rooms
- Updating room details
- Removing room records
- Modifying room prices
- Updating room amenities
- Uploading room images
- Managing room availability status
- Monitoring occupancy schedules
- Coordinating maintenance schedules

This information is stored in the Room entity and displayed to users when browsing accommodations through the online reservation system.

The admin can also update room information whenever changes occur. For example, room rates may be adjusted during peak seasons, amenities may be upgraded, or room descriptions may need modifications to reflect new services and accommodations offered by the resort.

The Manage Room Availability feature also allows administrators to remove inactive or unavailable rooms from the system when necessary. This helps maintain accurate room listings and prevents customers from attempting to reserve accommodations that are no longer operational.

One of the most important functions of this feature is updating room availability status in real time. The system continuously synchronizes room status based on reservation activities, occupancy records, cancellations, and maintenance schedules.

When a customer successfully books a room, the system automatically updates the room status from available to reserved. Upon guest check-in, the room status may change to occupied. After check-out and room preparation, the status returns to available.

If a reservation is canceled, the system immediately updates the room's availability status and reopens the room for future reservations. This real-time updating process helps prevent double bookings and scheduling conflicts.

The admin may also manually update room statuses when needed. For example, rooms undergoing maintenance, repairs, or cleaning may temporarily be marked as under maintenance or unavailable to prevent customer reservations during maintenance periods.

The Manage Room Availability feature is closely connected to several entities including Room, Reservation, Availability Schedule, Room Maintenance, and Admin. These entities work together to ensure that room occupancy and reservation schedules remain accurate and properly coordinated.

The feature significantly improves operational efficiency because administrators can monitor all accommodations through a centralized management system rather than relying on manual tracking methods. Automated availability updates reduce human error and improve reservation accuracy.

From the customer's perspective, real-time room availability improves the booking experience because users can immediately identify which rooms are available during their preferred dates. Accurate room information also helps customers make better reservation decisions.

Additionally, room availability management supports business planning and occupancy monitoring. Administrators can analyze room utilization, identify high-demand accommodations, monitor vacant rooms, and adjust operational strategies accordingly.

The system may also generate reports related to room occupancy rates, room availability trends, maintenance schedules, and reservation statistics. These reports help management evaluate resort performance and improve accommodation management.

Security and accountability are also supported because only authorized administrators can modify room records and availability statuses. Administrative actions related to room management may be recorded through the Audit Logs entity for monitoring and tracking purposes.

Overall, the Manage Room Availability feature plays a critical role in ensuring accurate reservation processing, organized accommodation management, real-time scheduling, and efficient resort operations within the Online Reservation and Management System for Messiah Inland Resort. Through effective room availability management, the system helps improve customer satisfaction, operational productivity, and reservation reliability.

- **Manage Reservations** – In the proposed Online Reservation and Management System for Messiah Inland Resort, the Manage Reservations feature enables administrators and authorized staff members to view, monitor, update, confirm, or cancel reservations based on resort policies, operational requirements, and room availability. This feature is essential because it helps maintain organized reservation processing, accurate booking records, and efficient accommodation management within the resort.

Reservation management is one of the core functions of the system because it ensures that all booking transactions are properly monitored and coordinated. Through this feature, administrators and staff can supervise reservation activities in real time and respond quickly to customer requests, schedule changes, and operational concerns.

The Manage Reservations feature is accessible through the administrative dashboard, where all reservation records are displayed and organized according to booking status, reservation dates, customer information, and room assignments. The system allows administrators and staff to perform several reservation management tasks including:

- Viewing reservation details
- Confirming reservations
- Updating booking information
- Modifying room assignments
- Monitoring reservation status
- Canceling reservations
- Tracking customer bookings
- Reviewing payment status
- Coordinating room schedules
-

When a customer submits a reservation request, the system stores the booking information within the Reservation entity. Reservation details may include:

- Reservation reference number
- Customer name and contact details
- Selected room or facility
- Check-in and check-out dates
- Number of guests

- Reservation status
- Payment information
- Special requests or notes

Administrators and staff can review these details to verify booking accuracy and ensure that reservations comply with resort policies and room availability schedules.

The reservation management process begins with monitoring incoming booking requests. Reservations may initially appear with a pending status until reviewed and confirmed by the administrator or authorized staff member.

After verifying room availability and booking information, the administrator may approve or confirm the reservation. Once approved, the reservation status is updated to confirmed, and the corresponding room availability is automatically adjusted within the Availability Schedule entity to prevent double bookings.

This flexibility helps accommodate customer requests and supports better customer service.

In cases where reservations need to be canceled, administrators and staff can process cancellations according to resort policies. Cancellations may occur because of customer requests, payment issues, scheduling conflicts, unavailable accommodations, or operational concerns.

When a reservation is canceled, the system automatically updates the reservation status and room availability in real time. The canceled room becomes available again for future bookings, ensuring efficient room utilization and accurate scheduling.

The Manage Reservations feature is closely connected to several entities including Reservation, Room, User, Payments, Availability Schedule, Notifications, and Admin.

These entities work together to ensure accurate reservation processing and organized booking management.

Notifications are also integrated into the reservation management process. Customers may receive automated messages regarding reservation confirmations, updates, cancellations, payment reminders, or schedule changes through email, SMS, or system notifications.

From the operational perspective, reservation management significantly improves efficiency because resort staff no longer need to manually organize reservation logs or booking schedules. Automated reservation tracking reduces human errors, improves accuracy, and saves administrative time.

The feature also improves customer satisfaction because booking concerns and reservation requests can be addressed more quickly and efficiently. Customers can receive timely updates regarding their reservations and enjoy more organized booking services.

Additionally, reservation management supports business monitoring and reporting. Administrators can generate reports related to booking trends, occupancy rates, cancellation statistics, and customer activities to help management evaluate resort performance and make informed decisions.

Security and accountability are also maintained because only authorized administrators and staff members can manage reservation records. Administrative actions related to reservations may be tracked through the Audit Logs entity to monitor booking activities and system changes.

The Manage Reservations feature plays an important role in maintaining smooth resort operations, accurate scheduling, and organized customer service processes. By centralizing reservation management within the system, the resort can provide more reliable, efficient, and professional reservation services.

Overall, the Manage Reservations feature is essential to the successful operation of the Online Reservation and Management System for Messiah Inland Resort. It supports accurate booking management, real-time reservation monitoring, operational efficiency, and improved customer satisfaction while ensuring that resort policies and accommodation schedules are properly maintained.

- **User Registration and Login** – In the proposed Online Reservation and Management System for Messiah Inland Resort, the User Registration and Login feature allows guests and customers to create personal accounts, securely log into the system, and access various reservation and account management functions. This feature is essential because it provides secure user identification, protects customer information, and allows personalized access to the system's services and functionalities.

The User Registration and Login process serves as the foundation of user interaction within the online reservation system. Through this feature, customers can manage reservations, monitor booking history, receive notifications, submit feedback, and access other resort services using their individual accounts.

The registration process begins when a guest creates a new account in the system. During registration, the user is required to provide important personal information such as:

- Email address
- Code

Once the required information is submitted, the system validates the entered data to ensure accuracy and completeness. The system may also verify whether the username or email address already exists to prevent duplicate accounts.

After successful registration, the system creates a User entity record in the database and securely stores the customer's information. Passwords are encrypted or protected using security measures to help prevent unauthorized access and improve data security.

The login process allows registered users to access their accounts using their username, email address, and password. During login, the system verifies the entered credentials against the stored user records before granting access to the system features.

If the login credentials are correct, the user is successfully authenticated and redirected to their account dashboard or homepage. Through their account, users can:

- Browse available rooms and facilities
- Check room availability
- Create reservations
- View reservation history
- Monitor booking status
- Receive notifications and confirmations
- Update personal information
- Submit customer feedback
- Manage cancellations or booking requests

If incorrect login credentials are entered, the system may display an error message and deny account access to maintain account security. The system may also implement additional security features such as password recovery, account verification, or login attempt monitoring.

The User Registration and Login feature is closely connected to several entities including User, Reservation, Notifications, Customer Feedback, Reservation History, and Audit Logs. These entities rely on authenticated user accounts to ensure that customer transactions and activities are properly recorded and managed.

The feature also improves security and accountability within the system because each reservation, payment transaction, feedback submission, or booking activity is linked to a specific registered user account. This helps administrators monitor customer activities and maintain accurate system records.

For customers, account registration improves convenience because personal information and reservation records are stored within the system. Users do not need to repeatedly enter the same details whenever making new reservations or inquiries.

The login feature also enhances customer experience by providing personalized access to reservation services. Customers can easily monitor their bookings, check reservation updates, and manage transactions through their accounts anytime and anywhere.

From the administrative perspective, user registration allows the resort to maintain organized customer records and improve communication with guests. Administrators can verify reservation ownership, send notifications, and provide customer support more effectively.

Additionally, user accounts support marketing and customer relationship management because the resort can analyze customer preferences, booking behavior, and reservation history to improve services and promotional strategies.

The User Registration and Login feature also contributes to system security by restricting access to authorized users only. Sensitive information such as reservation details, payment records, and customer data remain protected through authentication and account verification processes.

Audit Logs may also record login activities, failed login attempts, account updates, and user actions within the system. This improves system monitoring, security management, and accountability.

Overall, the User Registration and Login feature plays a critical role in ensuring secure system access, organized customer management, personalized reservation services, and efficient user interaction within the Online Reservation and Management System for Messiah Inland Resort. By providing secure account registration and authentication processes, the system improves customer convenience, data protection, and overall reservation management efficiency.

- **Receive Booking Confirmation** – In the proposed Online Reservation and Management System for Messiah Inland Resort, the Receive Booking Confirmation feature automatically sends confirmation details to guests after a successful reservation transaction. This feature is important because it provides customers with official verification that their booking has been successfully processed, recorded, and approved within the system.

Booking confirmations serve as proof of reservation and help ensure that both the customer and the resort maintain accurate and organized reservation records. By automatically sending confirmation details, the system improves communication, customer confidence, and reservation transparency.

The booking confirmation process begins once a user successfully completes a reservation request and the booking is verified by the system or approved by the administrator. After confirmation, the system automatically generates a reservation confirmation linked to the customer's reservation record and account information.

The system may deliver confirmation details through multiple communication channels including:

- Email notifications
- SMS messages
- In-system notifications

This flexibility ensures that guests can easily access their reservation details anytime for reference and verification purposes.

The Receive Booking Confirmation feature is closely connected to several entities including Reservation, User, Notifications, Payments, and Admin. These entities work together to ensure that reservation confirmations are accurate, updated, and properly delivered to customers.

Automated confirmation messages significantly improve operational efficiency because resort staff no longer need to manually contact customers for every reservation transaction. The system automatically processes and delivers booking confirmations in real time once reservations are successfully completed.

For customers, receiving an immediate booking confirmation improves trust and confidence in the reservation process. Guests can quickly verify that their selected room, reservation dates, and booking information were correctly processed by the system.

Confirmation messages also reduce misunderstandings and reservation disputes because customers can review all reservation details immediately after booking. If incorrect information is identified, guests can contact the resort or request updates before their scheduled stay.

The feature also supports better customer preparation because confirmation messages may include important reminders regarding:

- Check-in and check-out schedules
- Payment instructions
- Resort policies
- Cancellation rules
- Facility guidelines
- Contact information for inquiries or support

In cases where reservation details are updated, modified, or canceled, the system may automatically send updated notifications to inform the customer of any changes in their booking status.

From the administrative perspective, booking confirmations help maintain organized reservation records and improve reservation monitoring. Administrators can review confirmation logs, verify successful transactions, and monitor customer communications through the system dashboard.

The Receive Booking Confirmation feature also improves customer service because guests receive timely updates without needing to repeatedly contact resort staff for reservation verification. This reduces customer inquiries and helps streamline resort operations.

The feature contributes to system reliability and professionalism because automated confirmations create a more organized and efficient reservation process. Customers are more likely to trust a system that provides immediate and accurate booking verification.

Security and accountability are also enhanced because confirmation details are linked to authenticated user accounts and recorded reservation transactions. Notification activities may also be monitored through the Audit Logs entity for tracking and system monitoring purposes.

Overall, the Receive Booking Confirmation feature plays an important role in improving communication, reservation accuracy, customer satisfaction, and operational efficiency within the Online Reservation and Management System for Messiah Inland Resort. By automatically sending booking confirmations after successful reservations, the system ensures organized reservation management and provides guests with reliable booking verification and support.

- **Manage Customer Records** – In the proposed Online Reservation and Management System for Messiah Inland Resort, the Manage Customer Records feature allows administrators to store, organize, update, retrieve, and monitor guest information and booking history efficiently. This feature is essential because it helps maintain accurate customer records, improves reservation management, and supports better customer service within the resort

Customer records management plays an important role in ensuring that all guest information and reservation transactions are properly documented and securely stored in the system database. Through this feature, administrators can easily access customer details whenever needed for reservation processing, communication, reporting, and operational monitoring.

The Manage Customer Records feature is accessible through the administrative dashboard, where authorized administrators and staff can view and manage customer information in a centralized system. This eliminates the need for manual record keeping and improves data organization and retrieval.

The system stores important customer information such as:

- First name
- Last name
- Contact number
- Email address

These records are stored within the User entity and linked to other related entities such as Reservation, Payments, Reservation History, Notifications, and Customer Feedback.

The administrator can add new customer records whenever guests create accounts or make reservations through the system. Customer information is automatically saved in the database during the registration and booking process.

The retrieval function allows administrators to quickly search and access customer records using different search criteria such as:

- Customer name
- Contact number
- Email address
- Check In
- Check out

Fast and organized retrieval of customer information improves operational efficiency because administrators can quickly respond to inquiries, verify bookings, and resolve customer concerns without manually searching through paper records.

The feature also helps administrators handle customer concerns more effectively. If guests request reservation updates, cancellations, refunds, or account assistance, administrators can quickly retrieve the necessary information and provide proper support.

Customer record management is closely connected to the Reservation entity because reservation transactions are directly linked to specific customer accounts. This relationship helps maintain organized booking records and improves reservation tracking accuracy.

The Manage Customer Records feature also supports reporting and analytics functions. Administrators can generate reports regarding customer activity, frequent guests, reservation statistics, booking trends, and customer demographics to assist in business analysis and decision-making.

Security and privacy are important aspects of customer records management. The system ensures that sensitive customer information is protected from unauthorized access through secure login authentication, administrative access control, and database security measures.

Only authorized administrators and staff members are allowed to access or modify customer records. Additionally, activities related to customer record management may be monitored through the Audit Logs entity to improve accountability and system security.

The feature also contributes to better communication between the resort and its customers. Administrators can use stored customer information to send booking confirmations, promotional offers, reservation reminders, payment updates, and important announcements.

From the customer's perspective, organized record management improves convenience because reservation details, preferences, and account information are securely stored for future transactions. Returning customers can enjoy faster booking processes and more personalized services.

Overall, the Manage Customer Records feature plays an important role in improving customer management, reservation tracking, operational efficiency, and service quality within the Online Reservation and Management System for Messiah Inland Resort. By maintaining accurate and organized guest records, the system supports efficient resort operations, better customer relationships, and reliable reservation management.

- **Generate Reports** – In the proposed Online Reservation and Management System for Messiah Inland Resort, the Generate Reports feature enables the system to automatically produce reservation, operational, financial, and customer-related reports for monitoring, analysis, and decision-making purposes. This feature is important because it helps administrators and resort management organize information, monitor business performance, and evaluate resort operations more effectively.

The report generation process allows the system to collect, process, summarize, and present data gathered from different entities within the reservation system. These reports provide valuable insights regarding reservation activities, room occupancy, customer behavior, financial transactions, and operational performance.

The Generate Reports feature is essential because it transforms raw data into organized and meaningful information that supports management decisions and operational planning. Through automated reporting, administrators no longer need to manually organize reservation records or compute operational statistics, which helps reduce errors and save time.

Reservation reports contain detailed information regarding customer bookings such as reservation dates, room assignments, booking status, guest information, and payment details. These reports help administrators monitor current and upcoming reservations more effectively.

Booking summary reports provide an overview of reservation activities within a specific period such as daily, weekly, monthly, or yearly bookings. This allows management to track booking performance and identify trends in customer demand.

Occupancy reports monitor how frequently rooms and facilities are reserved and occupied. These reports help management determine occupancy rates, identify high-demand accommodations, and evaluate room utilization efficiency.

Revenue and payment reports provide financial summaries related to reservation transactions, payments received, unpaid balances, refunds, and resort income. These reports support accounting processes, budgeting activities, and financial monitoring.

Customer activity reports help administrators analyze guest behavior such as reservation frequency, preferred accommodations, booking history, and customer retention patterns. This information may help improve customer service and marketing strategies.

Cancellation reports provide details regarding canceled reservations, cancellation reasons, refund activities, and booking adjustments. These reports help management evaluate reservation policies and identify operational concerns affecting customer bookings.

Room availability reports allow administrators to monitor room status, vacant accommodations, maintenance schedules, and future booking availability. This improves room scheduling and accommodation management.

Facility usage reports summarize how frequently resort facilities such as swimming pools, cottages, event spaces, and recreational areas are being used by customers. These reports help management monitor facility demand and improve operational planning.

Customer feedback reports collect and summarize guest reviews, ratings, comments, and suggestions. These reports help management evaluate customer satisfaction and identify areas requiring service improvement.

Peak season analysis reports help administrators identify periods with increased reservation activity such as holidays, weekends, vacation seasons, or special events. This information supports staffing preparation, room allocation, and promotional planning during high-demand periods.

The Generate Reports feature is closely connected to several entities including Reservation, User, Room, Payments, Facilities, Customer Feedback, Reservation History, and Availability Schedule. Data gathered from these entities is processed and organized into accurate and updated reports.

The Generate Reports feature significantly improves operational efficiency because information can be accessed quickly without manually reviewing paper records or reservation logs. Automated report generation reduces administrative workload and improves decision-making speed.

The feature also supports strategic planning and business development. Management can use reports to evaluate resort performance, identify operational strengths and weaknesses, improve marketing strategies, monitor financial performance, and enhance customer services.

Security and accountability are also maintained because only authorized administrators and staff can access confidential operational and financial reports. Administrative activities related to report generation may also be monitored through the Audit Logs entity.

Additionally, report generation improves transparency and record keeping because all reservation transactions and operational activities are properly documented and organized within the system database.

From a managerial perspective, accurate reports provide essential information for evaluating business performance, monitoring operational efficiency, and supporting informed decision-making processes.

Overall, the Generate Reports feature plays a critical role in improving operational monitoring, reservation management, business analysis, and strategic planning within the Online Reservation and Management System for Messiah Inland Resort. Through automated and organized reporting functions, the system supports efficient resort operations, accurate record management, and informed administrative decision-making.

- **Handle Inquiries** – In the proposed Online Reservation and Management System for Messiah Inland Resort, the Handle Inquiries feature allows guests and customers to send questions, requests, and concerns regarding room availability, pricing, reservations, facilities, and resort services. This feature is important because it improves communication between the resort and its customers while providing guests with quick and convenient access to important information.

The Handle Inquiries feature supports customer service operations by allowing users to directly communicate with resort administrators or staff through the online system. Customers can submit inquiries anytime without needing to personally visit the resort or rely solely on phone calls for assistance.

Through this feature, guests may inquire about various resort-related concerns including:

- Room availability

- Room rates and pricing
- Reservation procedures
- Resort facilities and amenities
- Event spaces and cottages
- Check-in and check-out schedules
- Reservation policies
- Cancellation policies
- Special accommodations or requests
- Resort location and contact information

The inquiry process typically begins when a guest accesses the inquiry section within the system. Customers may be required to provide basic information such as:

- Full name
- Contact number or email address
- Subject of inquiry
- Message or concern details

The system then records the inquiry and forwards it to the administrator or authorized staff member responsible for customer support and communication.

The Handle Inquiries feature is closely connected to several entities including User, Reservation, Room, Facilities, Notifications, and Admin. These entities help administrators access relevant information needed to respond accurately to customer concerns and requests.

Administrators and staff can view incoming inquiries through the administrative dashboard, where messages are organized according to inquiry type, submission date, customer details, or response status. This organized inquiry management helps staff monitor customer concerns more efficiently.

Once an inquiry is reviewed, administrators or authorized staff can provide responses through:

- Email replies
- SMS notifications
- In-system messaging

This communication process helps ensure that customers receive timely and accurate information regarding their concerns or reservation requests.

The Handle Inquiries feature improves customer convenience because guests can quickly ask questions before making reservations. This helps customers make informed decisions regarding accommodations, pricing, facilities, and resort services.

The feature also supports reservation management because customers may inquire about available rooms, booking schedules, or reservation adjustments before completing transactions. Administrators can use inquiry information to assist customers more effectively and encourage successful bookings.

From the operational perspective, handling inquiries through the system improves communication efficiency because all customer messages are properly documented and organized within a centralized platform. Resort staff no longer need to manually track inquiries through paper records or scattered communication channels.

The feature also helps improve customer satisfaction because guests receive faster responses and more reliable support regarding their concerns and requests. Good communication contributes to better customer relationships and enhances the resort's professional image.

Additionally, inquiry records may help administrators identify frequently asked questions, common customer concerns, and areas where additional information or service improvements may be needed. This information may support operational improvements and customer service enhancements.

The Handle Inquiries feature may also integrate with the Notifications entity to send automated acknowledgment messages confirming that customer inquiries have been received and are being processed.

Security and accountability are also maintained because inquiry activities and responses may be recorded within the system database. Administrative actions related to customer inquiries may also be monitored through the Audit Logs entity.

From a business perspective, effective inquiry handling may help increase customer engagement and reservation opportunities. Guests who receive clear and timely responses are more likely to complete reservations and trust the resort's services.

The feature also supports marketing and customer relationship management because administrators can use inquiry interactions to provide promotional offers, recommend accommodations, or assist customers in selecting appropriate resort services.

Overall, the Handle Inquiries feature plays an important role in improving customer communication, service efficiency, reservation support, and customer satisfaction within the Online Reservation and Management System for Messiah Inland Resort. By providing a convenient platform for customer inquiries and support, the system helps strengthen customer relationships and improve the overall resort booking experience.

- **Update Reservation Status** – In the proposed Online Reservation and Management System for Messiah Inland Resort, the Update Reservation Status feature allows the system and authorized administrators to modify and monitor the status of reservations throughout the booking process. This feature is important because it helps maintain accurate reservation records, improve booking management, and ensure proper communication between the resort and its customers.

Reservation status updates provide real-time information regarding the progress and condition of customer bookings. By continuously updating reservation statuses, the system ensures that both administrators and guests are informed about the current state of reservations, room availability, and booking transactions.

The reservation process typically begins when a customer submits a booking request through the online reservation platform. At this stage, the system automatically assigns an initial status to the reservation, usually marked as pending while awaiting verification or administrative approval.

The system and administrators may update reservation statuses based on different booking activities, operational requirements, and customer actions. Common reservation statuses include:

- Pending
- Confirmed
- Cancelled
- Completed
- Checked-in
- Checked-out
- Rejected
- Expired

A pending status indicates that the reservation request has been successfully submitted but is still awaiting review, payment verification, or approval from the administrator.

Once the administrator verifies the reservation details and confirms room availability, the reservation status may be updated to confirmed. A confirmed reservation indicates that the booking has been officially accepted and recorded by the resort.

If a reservation cannot be accommodated due to unavailable rooms, incomplete information, payment issues, or policy violations, the administrator may update the reservation status to rejected or cancelled depending on the situation.

The cancelled status may also be applied when customers voluntarily cancel their reservations according to resort cancellation policies and system rules.

During the guest's stay, the reservation status may further change to checked-in once the customer arrives at the resort and occupies the reserved room. After the customer completes their stay and checks out, the reservation status may be updated to checked-out or completed.

The Update Reservation Status feature is closely connected to several entities including Reservation, User, Room, Availability Schedule, Notifications, Payments, and Admin. These entities work together to ensure accurate reservation processing and synchronized booking management.

Whenever a reservation status changes, the system may automatically update room availability in real time. For example:

- Confirmed reservations mark rooms as reserved
- Cancelled reservations reopen room availability

- Checked-in reservations mark rooms as occupied
- Completed reservations release rooms for future bookings

This synchronization helps prevent double bookings, scheduling conflicts, and inaccurate room availability information.

The feature also improves communication because customers receive notifications whenever reservation statuses are updated. Notifications may include:

- Reservation confirmations
- Cancellation notices
- Payment verification updates
- Booking reminders
- Check-in instructions
- Reservation modification alerts

These updates may be sent through email, SMS, or in-system notifications to ensure that customers remain informed throughout the reservation process.

From the administrative perspective, updating reservation statuses helps administrators monitor booking progress, verify transactions, manage room occupancy, and coordinate resort operations more effectively.

The feature also improves operational efficiency because reservation monitoring becomes centralized and automated. Administrators and staff can quickly identify pending reservations, confirmed bookings, cancellations, and completed transactions through the system dashboard.

The Update Reservation Status feature supports accountability and transparency because all booking activities and status changes are properly recorded within the system database. Reservation updates may also be tracked through the Audit Logs entity to monitor administrative actions and system activities.

In addition, reservation status management helps improve customer satisfaction because guests receive accurate and timely information regarding their bookings. Customers can easily track the progress of their reservations without repeatedly contacting resort staff for updates.

The feature also contributes to better reporting and analytics because reservation statuses provide important data regarding booking trends, cancellation rates, occupancy levels, and operational performance.

Security is maintained because only authorized administrators and system functions are allowed to modify reservation statuses. This prevents unauthorized changes and protects the integrity of reservation records.

Overall, the Update Reservation Status feature plays a critical role in maintaining accurate booking management, real-time reservation monitoring, operational coordination, and customer communication within the Online Reservation and Management System for Messiah Inland Resort. By allowing the system and administrators to continuously monitor and update reservation conditions, the feature supports efficient resort operations and reliable customer service.

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education